

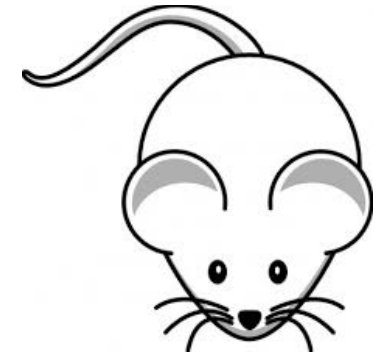
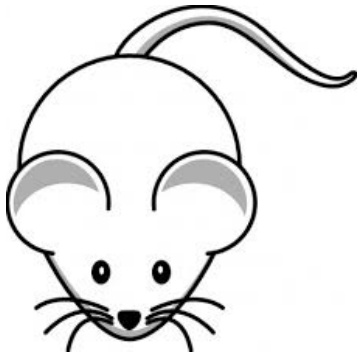


Animal experimentation or clinical research: ethical and regulated frameworks

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I. Experiment on a living individual:
a necessary ethical reflection

Ethics :

comes from the ancient Greek word êthikon = customs/morality

part of philosophy that considers the foundations of morality.

the set of moral principles that underlie someone's conduct

➤ **Ethics is the science of morality and customs.**

Moral :

refers to a set of values and principles

Difference between the two:

Ethics is not a set of values or principles in particular

This is **a reasoned reflection aimed at doing good.**

It is about **questioning the moral values and principles that should guide our actions** in different situations, with the aim of acting in accordance with them.

The ethical approach in science:

"a reflective approach to the values and purposes of scientific research" (Coutellec , 2019).

- questioning the reasons for the research and its possible consequences on human dignity
- reflection on the practices and choices adopted
- a dynamic approach that must be adapted throughout the research

ethics of life sciences = bioethics

several major fields that frame medical, scientific and environmental practices.

- **medical bioethics** : studies choices related to human health, including questions about the beginning and end of life.
- **research ethics** : sets the rules for testing new drugs or treatments, protects people who participate in scientific studies, ensures the honesty of researchers.
- **animal ethics** : focuses on the status of animals and sets limits on the use of animals in scientific research and industry.
- **environmental ethics** : examines the relationship between humans and nature, protects biodiversity, ecosystems and the climate from human actions.
- **neuroethics** : studies the moral issues related to the brain, analyzing technologies that allow to read or modify brain activity.
- **biotechnology ethics** : assesses the risks of technical innovations, genetically modified organisms (GMOs), human DNA modification, or synthetic biology.

Differences between ethics, professional conduct, and scientific integrity:

professional conduct: rules and obligations to be followed by a given profession

ethics: non-prescriptive, it draws on the values of researchers and their research discipline

scientific integrity: set of professional standards that guarantee the honest and scientifically rigorous nature of researchers' work

II. The framework for animal experimentation:
use of live animals for scientific purposes

Animal experimentation: a necessity

Currently, the scientific community agrees that animal studies are necessary :

- to understand the integrated functioning of an organism: physiology
- to understand how a disease develops: pathology
- to study the effects of a drug: pharmacology

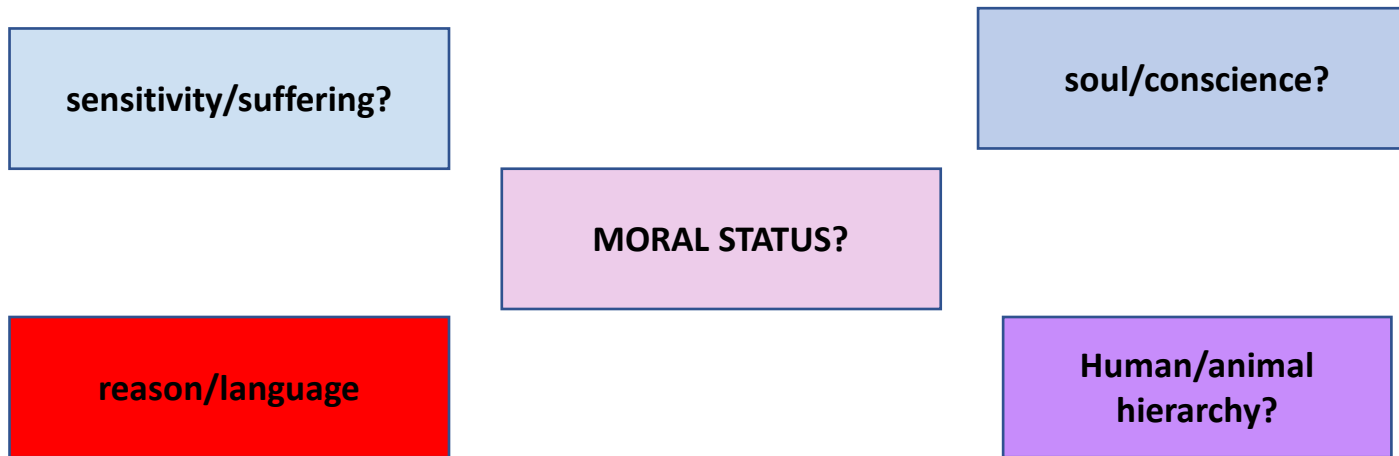
Among the arguments:

- Animals, and mammals in particular, have many things in common with humans.
- Most human diseases also exist in animals.

The implementation of this observation of necessity is based on several centuries of ethical reflection

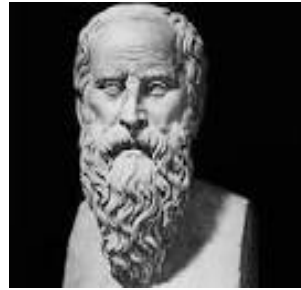
A little bit of history...

The moral status of animals in history: a long philosophical debate

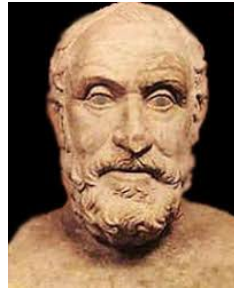




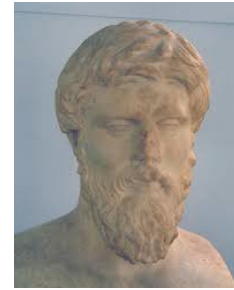
Pythagoras
6th century
BC



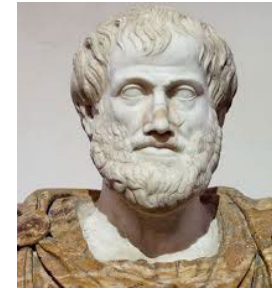
The Cynics
5th century
BC



The Skeptics
4th century
BC



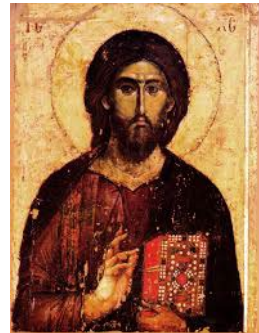
Plutarch
1st century



Aristotle
4th century



The Stoics
4th century



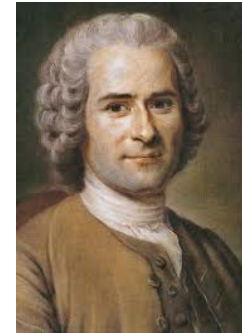
Christianity
Until the
Renaissance



Descartes
17th century



Kant
18th century



Rousseau
18th century



Bentham
18th century

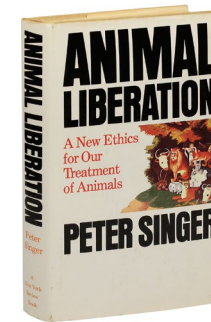


Claude Bernard - Inventor of experimental medicine - 19th century

"The physiologist is not a man of the world, he is a scholar, a man absorbed by a scientific idea that he pursues; he no longer hears the cries of animals, he no longer sees the flowing blood, he sees only his idea and perceives only organisms that hide from him problems that he wants to discover."

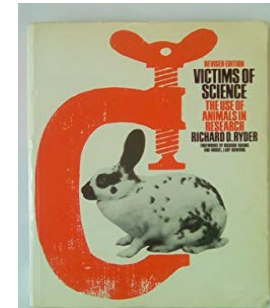
Welfarism P. Singer (1975)

Balanced utilitarianism: expected benefits will outweigh the suffering inflicted / animal welfare cost/benefit concept



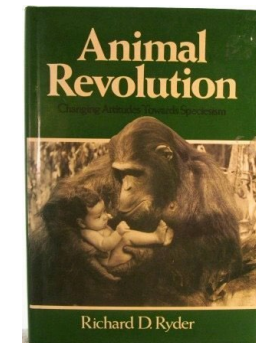
Abolitionism T. Reagan (1983)

The scientific enterprise has, in and of itself, no intrinsic value

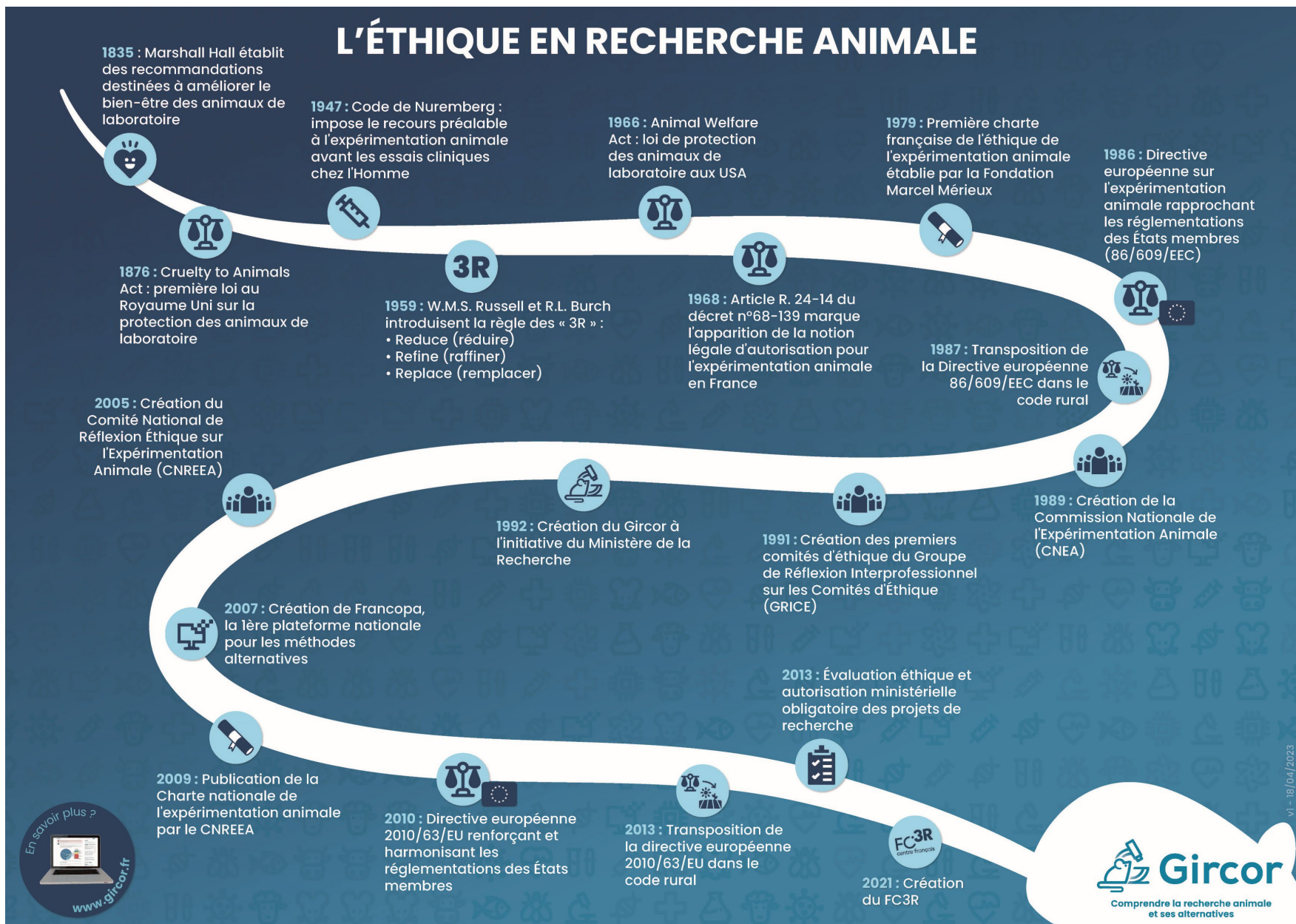


Anti- speciesism RD Ryder (1989)

Humans do not take precedence over other species, and equality is necessary. Anti-specists advocate for the importance of the individual, not the species to which they belong.



Ethical reflection has led to several regulations since the 19th century .



The use of live animals for scientific purposes today: figures on animal experimentation

In France :

around **2 million animals per year** are used for experimental purposes

Therefore : 0.03 animals/ year / citizen = **1 animal for every 35 citizens**

stable, or even slightly declining, over the last decade

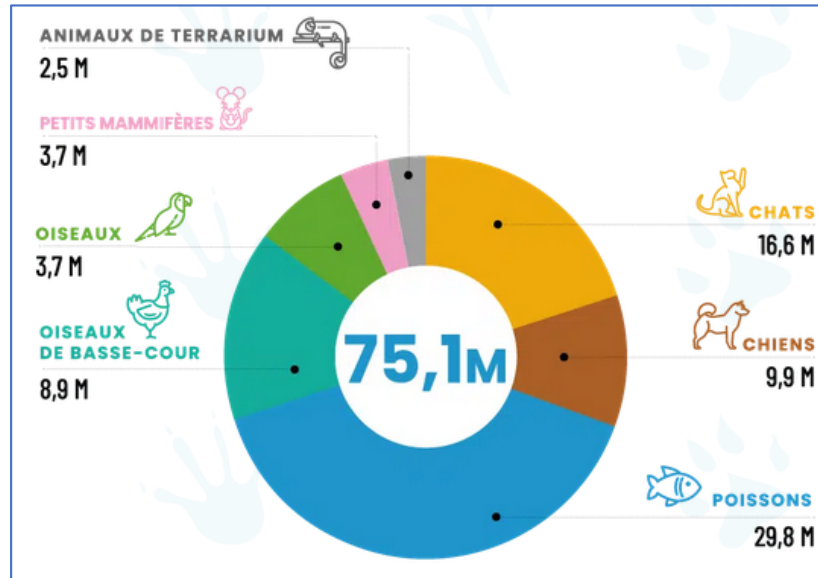
compare to :

☞ **75 million** pets (FACC0 -ODOXA data 2024)

☞ In 2020, **1.88 million tonnes of poultry** were consumed in France.

that's **28.5 kg of poultry per year per inhabitant**

☞ Rat population in Paris: estimated at between **4 and 8 million** individuals (i.e., 1.5 to 2 rats per inhabitant)



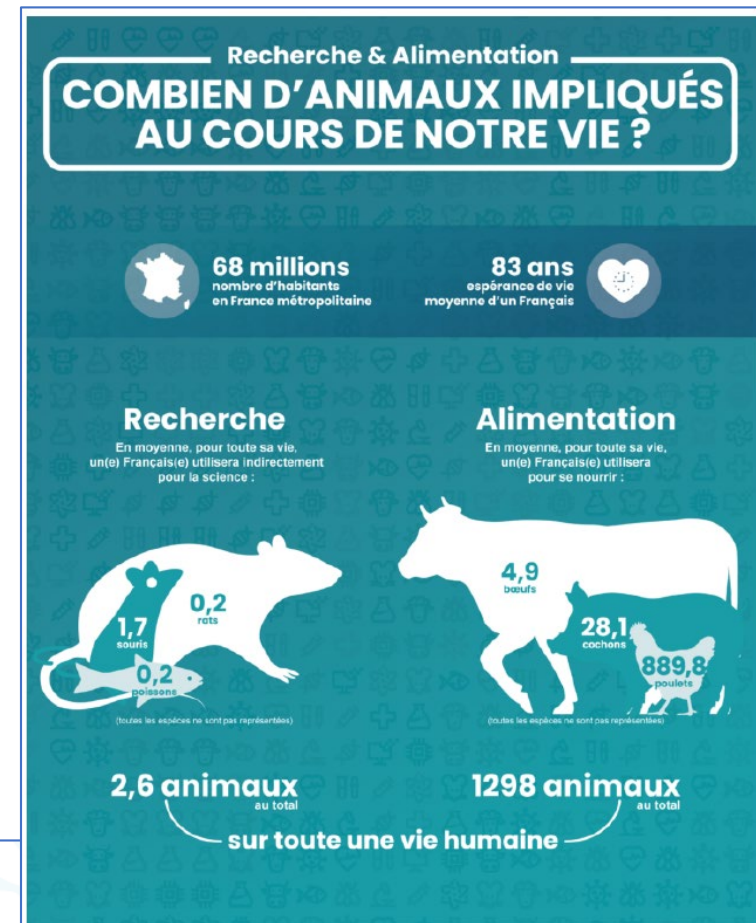
61% des Français possèdent des animaux de compagnie.



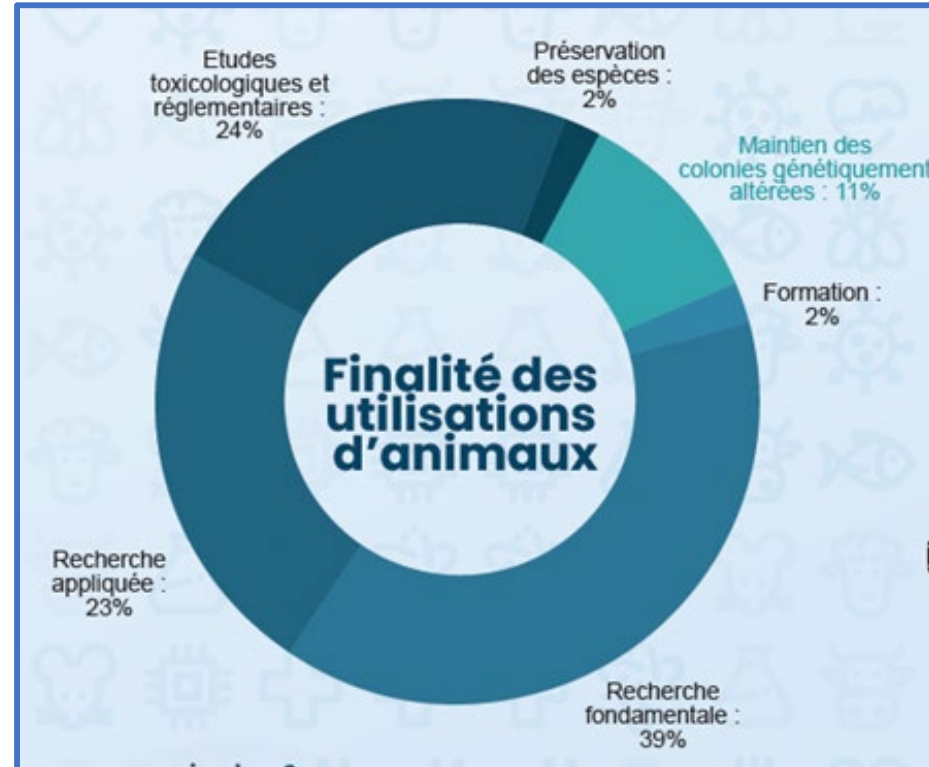
55% des Français ont un chien et/ou un chat ;
30% ont un chien et **39%** ont un chat.



Depuis 1976, le nombre total d'animaux de compagnie en France a été multiplié par 2,5 pour atteindre **75 millions**.



Animal research: what for?



MESRI data, 2022

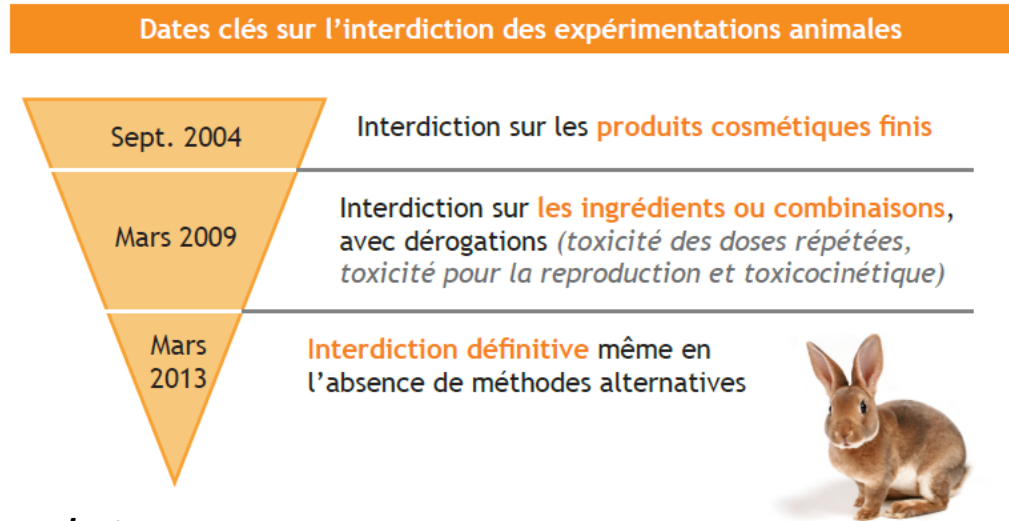
<https://www.recherche-animale.org>

Especially :

- 90% of veterinary medicines are identical or very similar to human medicines.
- vaccines and surgical techniques are first developed in animals,
- the differences between humans and animals allow for the development of new medicines.

Combating a misconception

Are animal tests allowed for cosmetic products? **NO**



Article 18 of the regulation on cosmetic products :

ban on the marketing of cosmetic products

- whose final formulation has been tested on animals
- containing ingredients or combinations of ingredients that have been tested on animals

ban on conducting animal experiments in the European Community to test

- finished cosmetic products ,
- the ingredients or combinations of ingredients

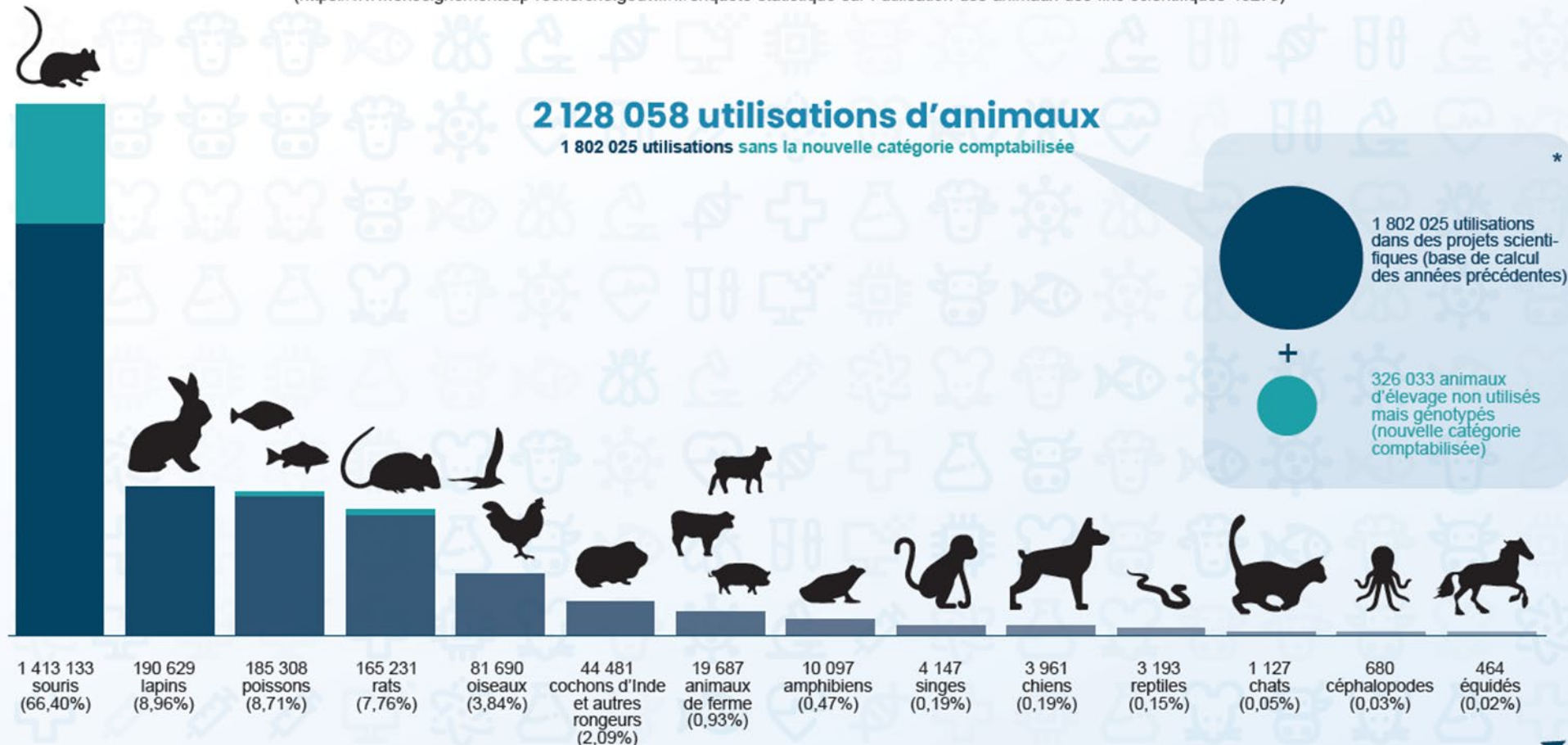
In exceptional circumstances , when there are serious concerns about the safety of an existing ingredient in a cosmetic product, a Member State may request the Commission to grant an exemption to the prohibitions.

Note: The majority of ingredients used in cosmetic products are also used in many other consumer and industrial products for which animal testing may be a requirement under the regulations in force applicable to the sector concerned.

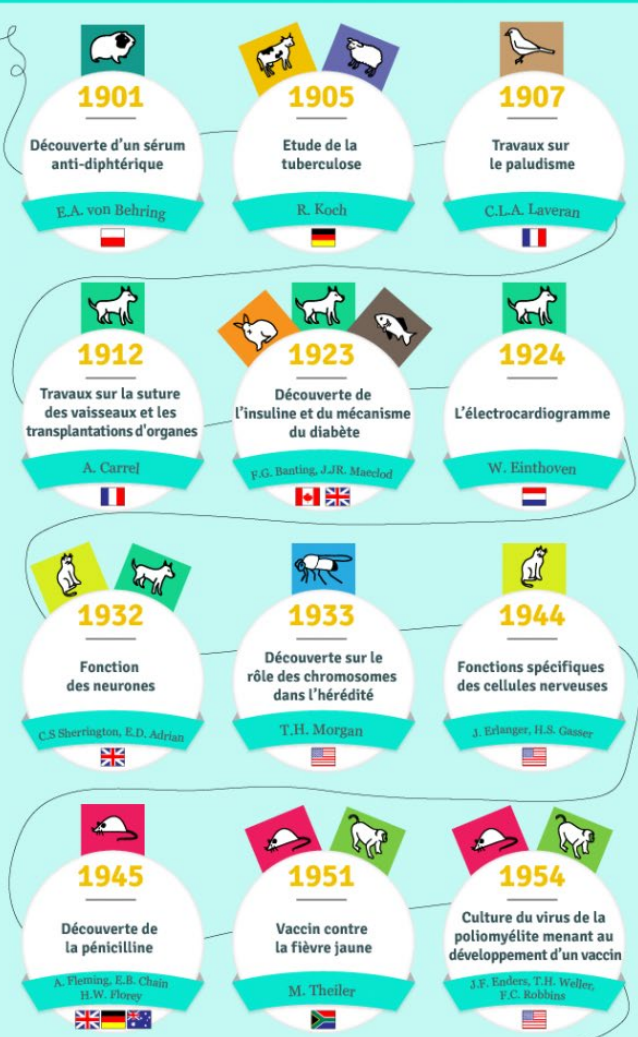
What species used in animal research ?

Utilisation des animaux à des fins scientifiques dans les établissements français – Statistiques 2022

Source : Enquête statistique 2022 : utilisation des animaux à des fins scientifiques - Ministère de l'enseignement supérieur, de la recherche
(<https://www.enseignementsup-recherche.gouv.fr/fr/enquete-statistique-sur-l-utilisation-des-animaux-des-fins-scientifiques-46270>)



Major biological and medical discoveries are linked to animal experimentation



1920: Use of insulin as a treatment for diabetes in **dogs** .

1930: Discovery of the effects of anesthesia on **laboratory rats** .

1940: Penicillin, as an antibiotic, was discovered through tests on **mice** .

1950: Development of hip prostheses following studies on **sheep** .

1960: Antidepressant effect of molecules demonstrated in **rats** .

1970: Manufacture of an inhaler against asthma after tests on **guinea pigs** .

1980: application of magnetic resonance imaging (MRI) as a diagnostic technique in **pigs** .

1990: Discovery of antiretroviral drugs against HIV in **monkeys** .

2000: Development of a vaccine against cervical cancer in **rabbits** .

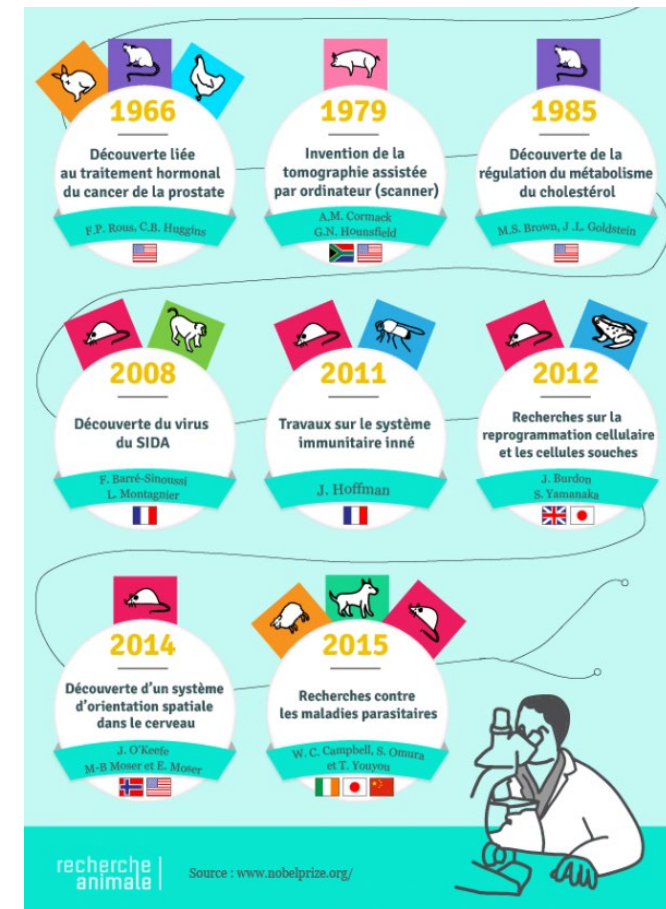
2010: Use of stem cells for cardiac tissue repair in **zebrafish**

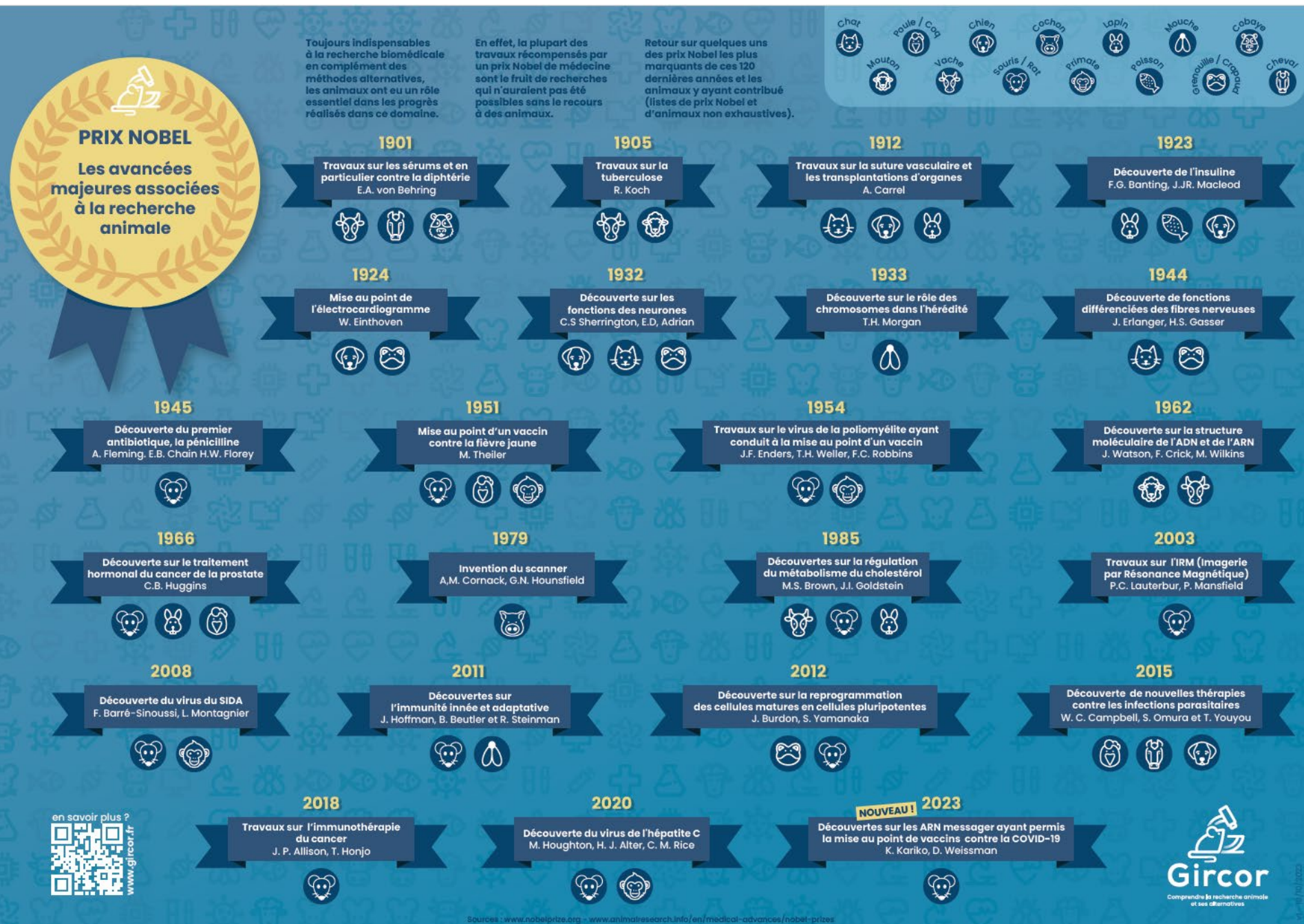
...

2020: SarsCov-2 pandemic!

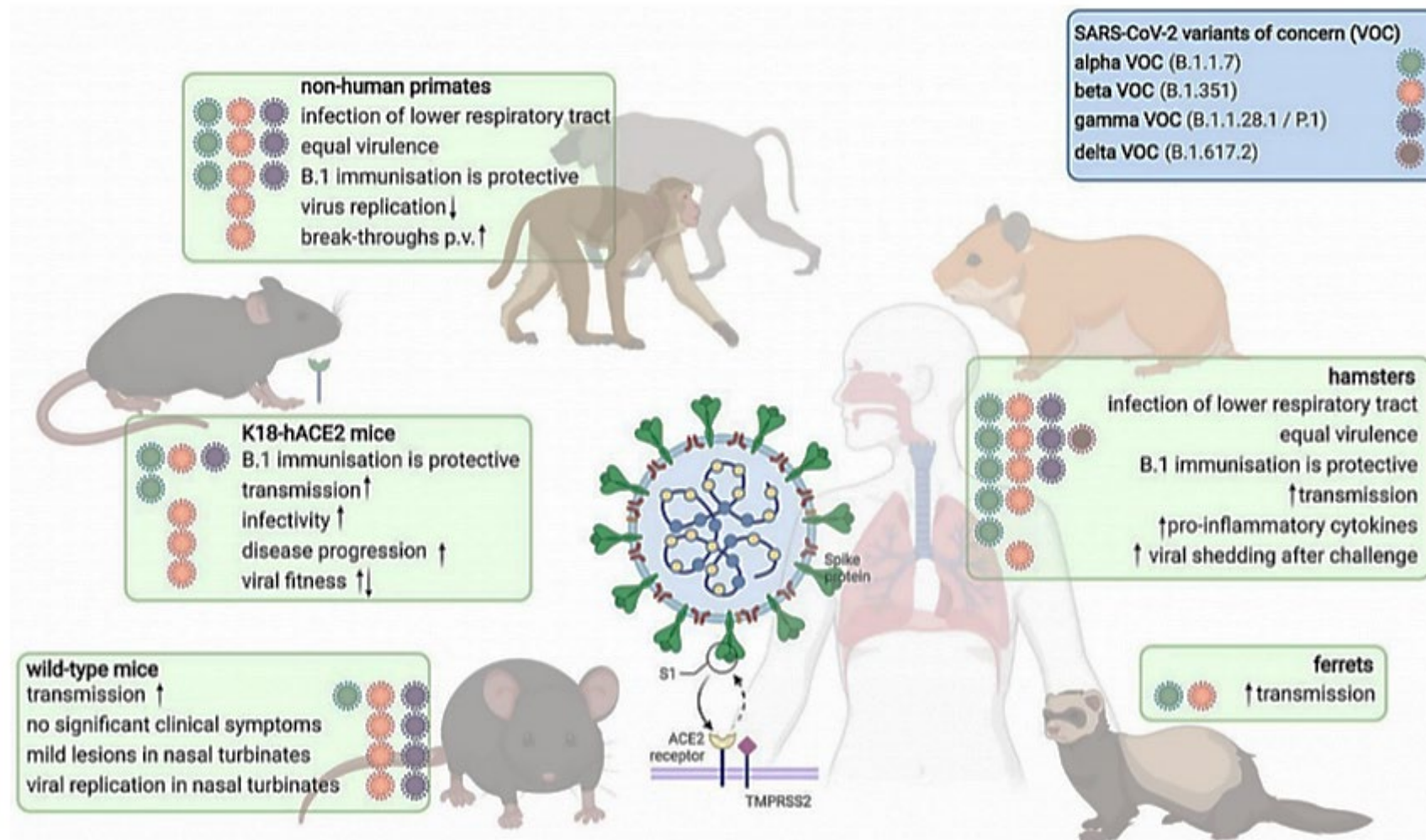
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2024: microRNA and **C. elegans**





The combination of animal models has enabled progress in research against COVID-19:

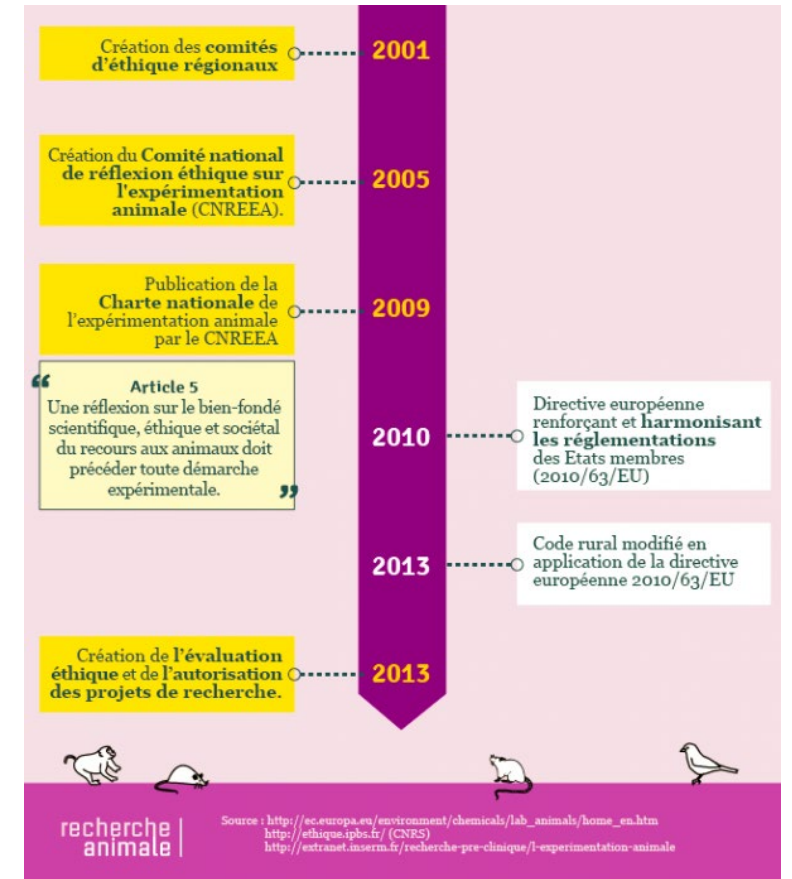
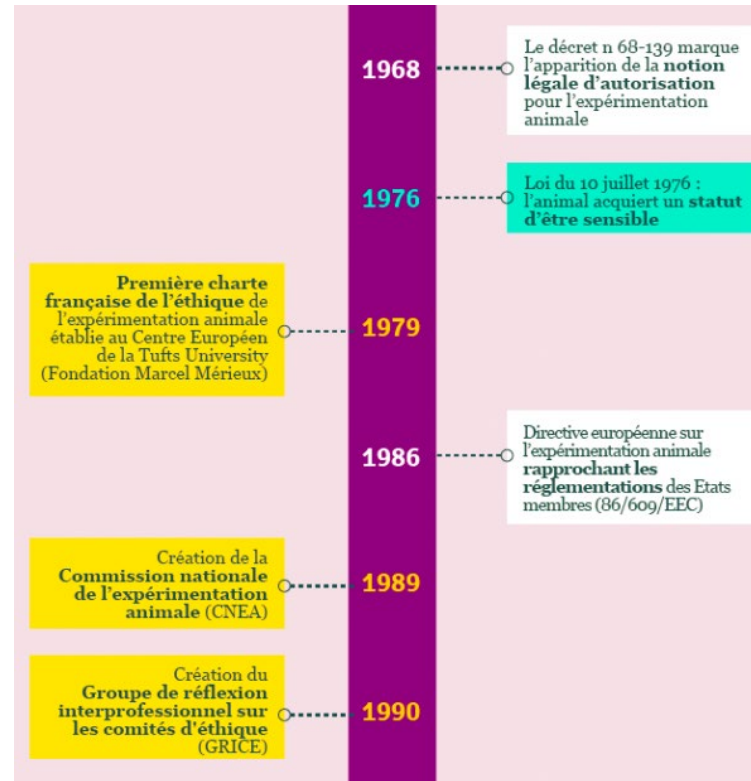
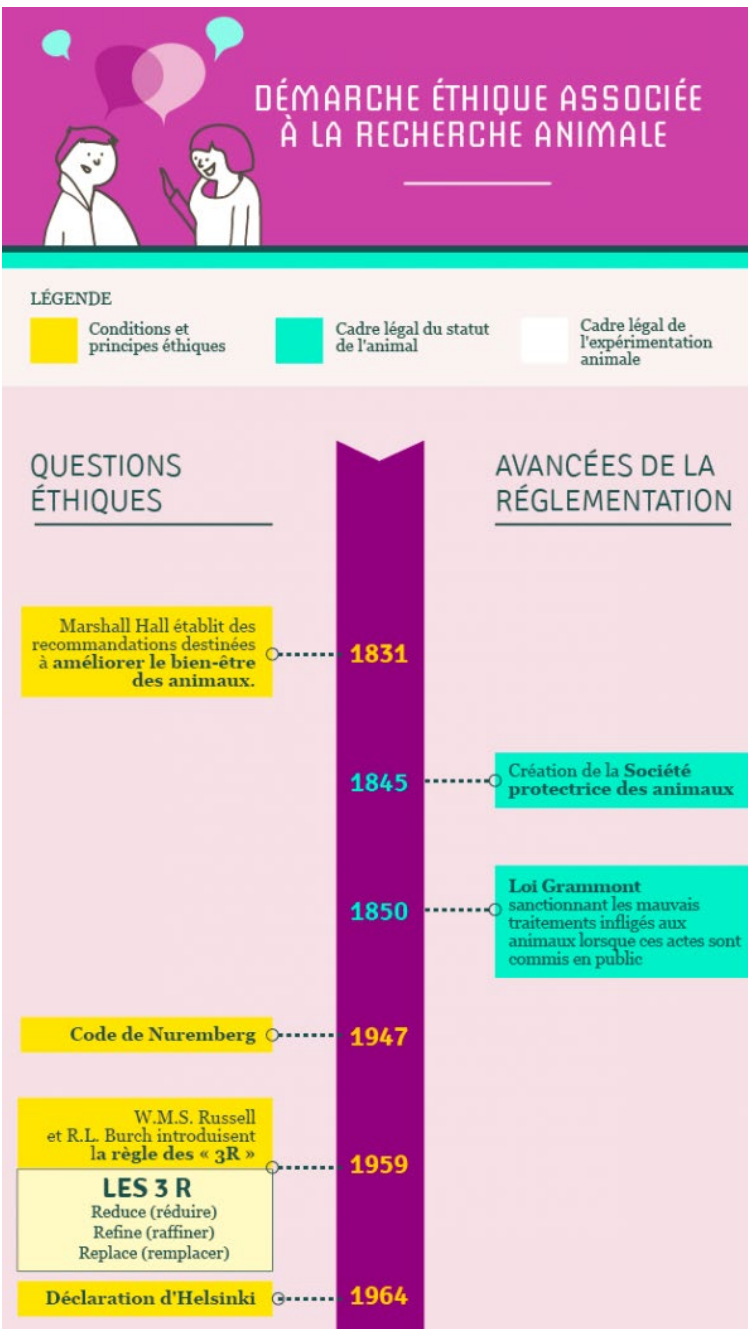


Détail des résultats en matière de virulence et de transmission des variants de la covid en fonction des modèles animaux. © CC0 Muñoz-Fontela et al.

Animal experimentation: a necessity, within an ethical framework

The necessity of animal experimentation has been the subject of philosophical and ethical debates within society for a long time.

This societal reflection has led to the development of an ethical framework and increasingly precise regulations.

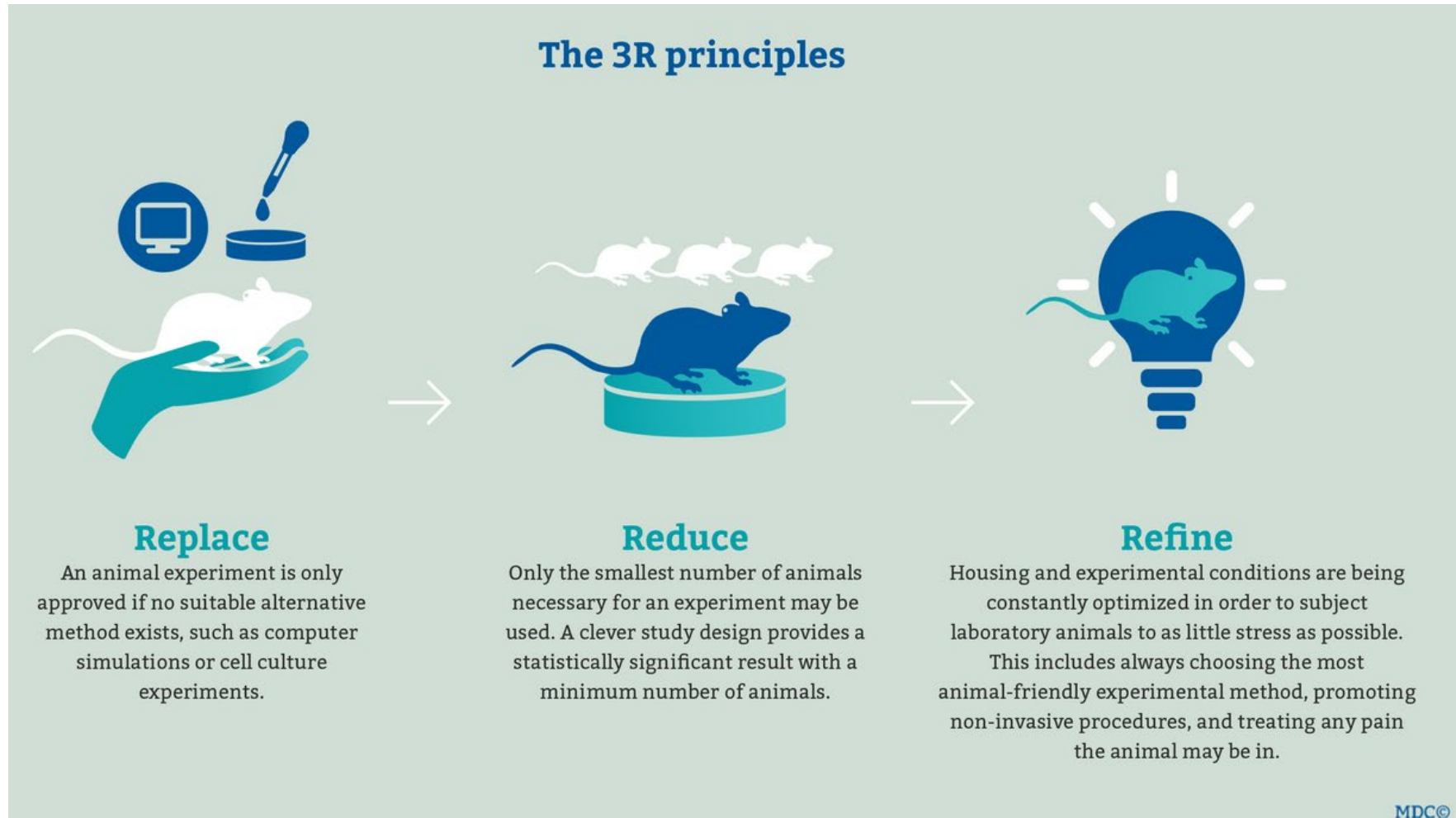


The central point of the ethical framework: the 3R principles

(Russell WMS & Burch RL (1959) "The Principles of Human Experimental Technique")

Animal models are **irreplaceable within a specific research framework** :

- biomedical research combines *in silico*, *in vitro*, *in vivo methods* and clinical trials.
each of them plays a different role in understanding biological mechanisms
- **animal testing is prohibited** if the expected information can be obtained by another method.



Animal experimentation: a necessity, within an ethical and in a **regulated framework**

European and French regulations :

- European Directive 2010/63/EU was reviewed in November 2017.
 - French law: Rural and Maritime Fishing Code, Article 6, L.214 and R214-87 to 130: use of live animals for scientific purposes
- Decree 2013-118 (February 1, 2013): protection of animals used for scientific purposes
+ 5 prescriptions supplemented by best practice guides
Decree amended on November 15, 2025



Arrêté du 15 novembre 2025 modifiant l'arrêté du 1er février 2013 fixant les conditions d'agrément, d'aménagement et de fonctionnement des établissements utilisateurs, éleveurs ou fournisseurs d'animaux utilisés à des fins scientifiques et leurs contrôles

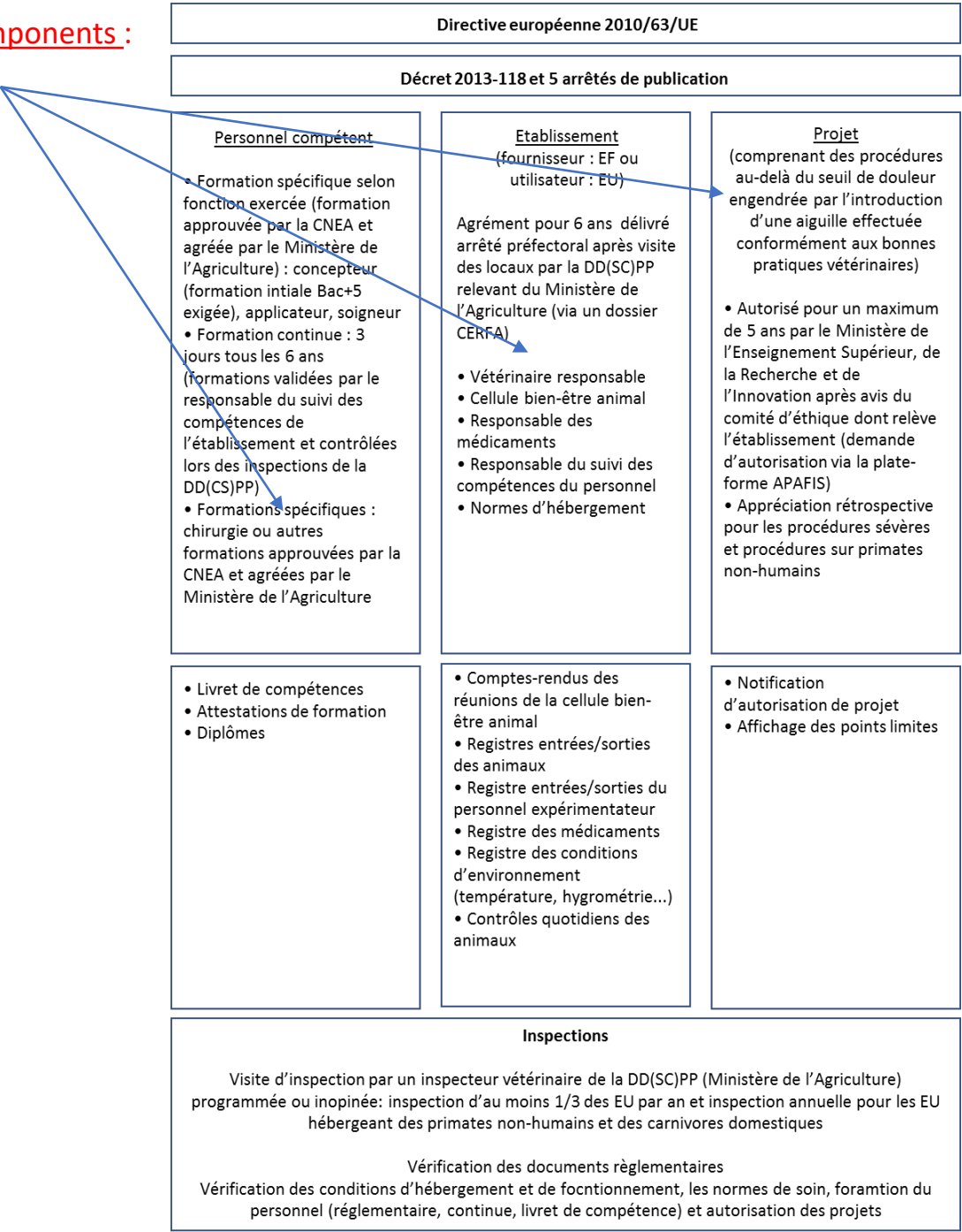
EN VIGUEUR AU 13/09/2026



The law requires:

- to train staff before working with animals,
- to approve and monitor animal suppliers, as well as breeding and experimentation facilities,
- to respect the principles of the 3Rs (replacement, reduction, refinement) and add a 4th R when possible: rehabilitation/decommissioning.
- to have the experimental protocols evaluated by an ethics committee and authorized by the Ministry of Research and Innovation **before the start of the experiments.**

The law has three main components :



Some details on the implementation of the law for laboratories and researchers...

Scope of application of the law

Animals concerned:

- live vertebrates, including autonomous larval or fetal forms of mammals, in the last third of their development or before the last third if suffering is likely to persist after this period.
- live cephalopods
- domestic or wild fauna



Experimentation is lawful if:

- necessary character (respect for the 3Rs)
- basic, translational or applied research for:
 - drug development (human or veterinary),
 - physiology,
 - food safety,
 - specialized higher education or vocational training
 - prevention, prophylaxis, treatment of diseases of humans, animals or plants.

The following are not within the scope of application :

- actions carried out on farms for non-experimental purposes
- acts carried out for recognized breeding purposes
- actions carried out for the purpose of identifying an animal
- practice of veterinary medicine for non-experimental purposes
- veterinary clinical trials required for marketing authorization of a veterinary medicinal product
- practices likely to cause pain, suffering, or distress LESS than that caused by the insertion of a needle performed according to good veterinary practice

The 3 main components of the law :

1. The staff

Directive européenne 2010/63/UE		
Décret 2013-118 et 5 arrêtés de publication		
<u>Personnel compétent</u> • Formation spécifique selon fonction exercée (formation approuvée par la CNEA et agréée par le Ministère de l'Agriculture) : concepteur (formation initiale Bac+5 exigée), applicateur, soigneur • Formation continue : 3 jours tous les 6 ans (formations validées par le responsable du suivi des compétences de l'établissement et contrôlées lors des inspections de la DD(SC)PP) • Formations spécifiques : chirurgie ou autres formations approuvées par la CNEA et agréées par le Ministère de l'Agriculture	<u>Etablissement</u> (fournisseur : EF ou utilisateur : EU) Agrément pour 6 ans délivré arrêté préfectoral après visite des locaux par la DD(SC)PP relevant du Ministère de l'Agriculture (via un dossier CERFA) • Vétérinaire responsable • Cellule bien-être animal • Responsable des médicaments • Responsable du suivi des compétences du personnel • Normes d'hébergement	<u>Projet</u> (comprenant des procédures au-delà du seuil de douleur engendrée par l'introduction d'une aiguille effectuée conformément aux bonnes pratiques vétérinaires) • Autorisé pour un maximum de 5 ans par le Ministère de l'Enseignement Supérieur, de la Recherche et de l'Innovation après avis du comité d'éthique dont relève l'établissement (demande d'autorisation via la plateforme APAFIS) • Appréciation rétrospective pour les procédures sévères et procédures sur primates non-humains
• Livret de compétences • Attestations de formation • Diplômes	• Comptes-rendus des réunions de la cellule bien-être animal • Registres entrées/sorties des animaux • Registre entrées/sorties du personnel expérimentateur • Registre des médicaments • Registre des conditions d'environnement (température, hygrométrie...) • Contrôles quotidiens des animaux	• Notification d'autorisation de projet • Affichage des points limites
Inspections Visite d'inspection par un inspecteur vétérinaire de la DD(SC)PP (Ministère de l'Agriculture) programmée ou inopinée: inspection d'au moins 1/3 des EU par an et inspection annuelle pour les EU hébergeant des primates non-humains et des carnivores domestiques Vérification des documents réglementaires Vérification des conditions d'hébergement et de fonctionnement, les normes de soin, formation du personnel (réglementaire, continue, livret de compétence) et autorisation des projets		

When we say "the staff", who are we talking about?

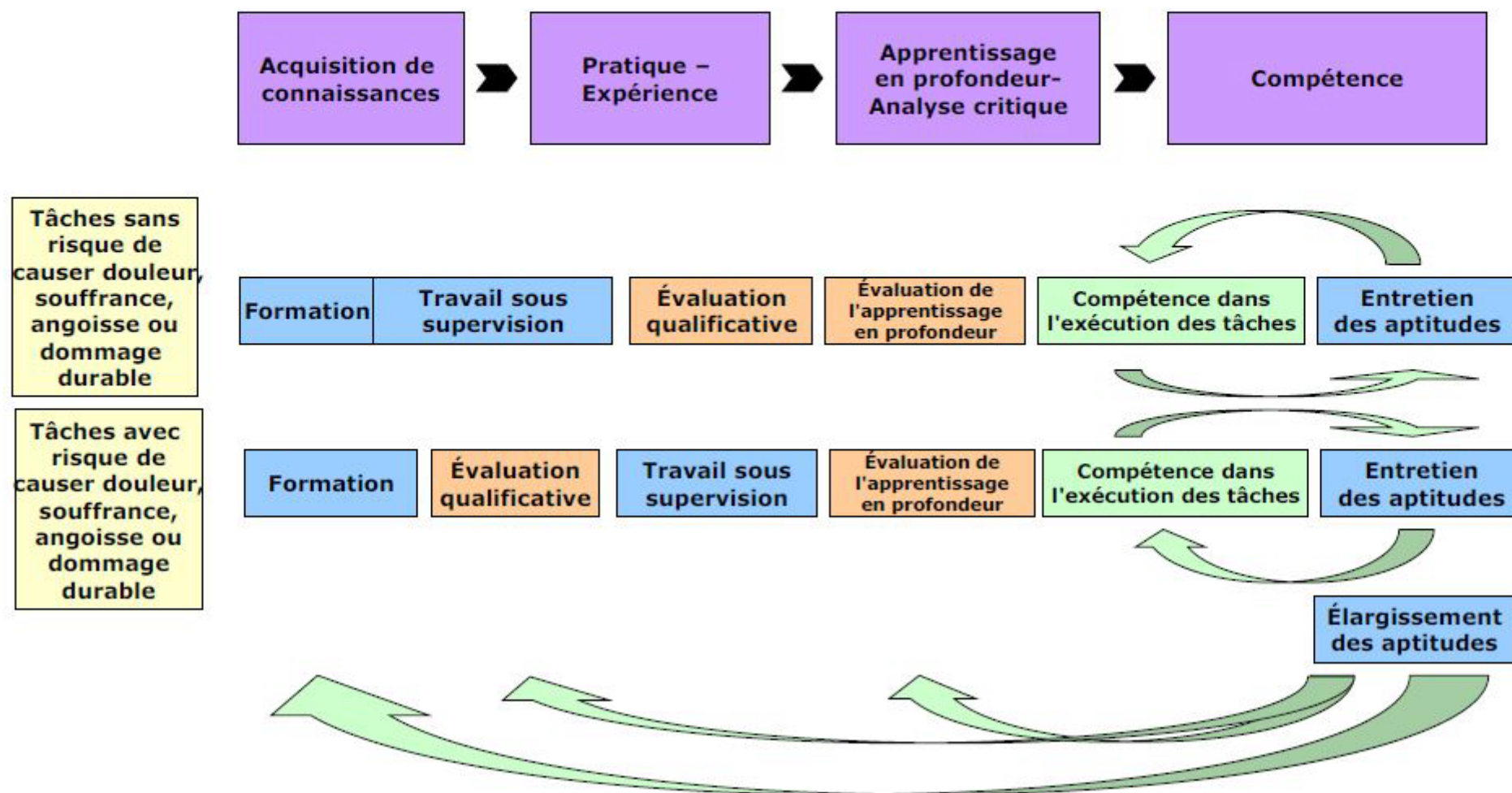
What the law says:

"Staff must be qualified and in sufficient numbers relative to the species housed and the studies carried out."

3 categories:

- designer : responsible for the design and implementation of experimental procedures
 - hold a degree equivalent to a 5-year university degree (bac+5).
 - regulatory training "designer" or "level 1"
 - surgical training if surgical procedures
- applicator : application and participation in experimental procedures
 - regulatory training "applicator" or "level 2"
 - surgical training if surgical procedures
- animal caretaker : animal care
 - regulatory training "animal caretaker" or "level 3"

PROCESSUS DE FORMATION AU TITRE DE LA DIRECTIVE 2010/63/UE



Initial regulatory training courses are defined by law in terms of training duration and training content.

Thèmes	Personnels appliquant des procédures expérimentales aux animaux	Personnels concevant des procédures expérimentales et des projets	Personnels assurant les soins aux animaux	Personnels assurant la mise à mort des animaux
Réglementation française applicable à l'expérimentation animale, dont les exigences de remplacement, de réduction et de raffinement				
Principes éthiques concernant les relations entre l'homme et l'animal, règle des 3 R Rôle et fonctionnement des comités d'éthique				
Procédures expérimentales faiblement invasives, sans anesthésie : théorie et pratique				
Méthodes alternatives				
Connaissances de base en biologie des espèces, dont la physiologie, l'anatomie, l'alimentation, reproduction, comportement, entretien et techniques d'enrichissement en rapport avec les caractéristiques physiologiques, la génétique et les modifications génétiques				
Reconnaissance des signes de détresse, de douleur et de souffrance propres aux espèces utilisées le plus couramment				
anesthésie et analgésie				
Recours aux points limites adaptés				
Méthodes d'euthanasie				
Gestion et suivi de la santé animale et de l'hygiène				
Équipements et matériels d'animalerie : description, utilisation, entretien				
Méthodes de transport, maniement, contention des animaux propres à chaque espèce				
Conception de procédures expérimentales et de projets				

[Order of 1 February 2013 relating to the acquisition and validation of staff skills](#)

Regulatory training duration	45 hours	57 hours	34 hours	must be designer or applicator
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Example of a skills booklet: that of the IBPS

LIVRET DE COMPETENCES

NOM :
PRENOM :

FORMATION INITIALE			
Intitulé du diplôme	Niveau d'études (BEP, Bac, Bac + n années ...)	Date d'obtention	Commentaires / Remarques

FORMATION SPECIFIQUE A L'EXPERIMENTATION ANIMALE					
Niveau / Fonction	Intitulé de la formation	Espèce concernée le cas échéant	N° agrément de la formation (suivie en France)	Date de la formation	Date validation de la formation suivie

FORMATION CONTINUE					
Intitulé de la formation suivie	Espèce concernée le cas échéant	Mode acquisition (formation pratique, théorique, séminaire, colloque ...)	Date de la formation	Durée de la formation (nb de jours)	Date validation de la formation suivie

Nb de jours formation continue : 0,00

COMPETENCES - VALIDATION DES ACQUIS					
Compétence acquise	Etablissement(s)	Espèce concernée le cas échéant	Nom tuteur(s)	Nom du Validateur	Date validation

DIVERS				
Fonctions exercées	Nom de l'établissement	N° d'enregistrement du Comité d'éthique	Dates d'exercice	Commentaires

essential obligations of staff



- all staff must have received training appropriate to their duties in the year following his appointment
- it is forbidden to perform surgical procedures without having completed the specific training.
(no one-year grace period in this case)
- staff must complete **3 days of continuing training** in animal experimentation **every 6 years.**
for all functions

These training courses must focus on animal experimentation: implementation of the 3Rs (Reduce, Reuse, Recycle), acquisition of technical skills (surgery is not a continuing education program)

no official validation by the Ministry of Agriculture but control during visits by the DDPP.

- the **correct execution of experimental procedures** must be validated by the training manager designated by the institution where the experiments take place.
- a **skills booklet** up to date. It must include the following information:
 - initial training (dates, institution that awarded the diploma)
 - specific training courses (dates, institution that awarded the diploma)
 - continuing education courses (dates, duration, program)
 - validation of technical skills by the tutor (date)

The experimenter must be able to present their skills booklet at any time while experimenting.

So what does this mean for a Master's (M1 or M2) internship ?

He is within the one-year grace period to acquire regulatory training (designer or applicator).

He must hold a skills booklet

He must work under supervision

before having validated his technical skills with the person in charge of the skills booklets at his animal facility

He is not authorized to perform surgical procedures.

Other regulatory staff required for the accreditation of an establishment

the veterinarian :

- advice on well-being
- orders the treatments for the animals

the structure of animal welfare (article 4.II of the decree of February 1 · 2013):

- advises staff on acquisition, accommodation, care, and the implementation of the 3Rs.
- ensures technical and scientific monitoring,
- drafts the BEA procedures
- oversees the placement of animals
- verifies the suitability of the project authorization to the project, ensures the progress and results of the projects on animals (documents the number of animals already entered into the project)

The BEA unit's monitoring documents must be kept for 5 years.

The person responsible for skills monitoring :

- keeps the skills tracking table up to date
- validates individual skills booklets: initial, continuing, and specific training

the pharmacy manager :

designer level required

- ensures supply and inventory management,
- supervises the use within the exclusive framework of the procedures

(Outside of the procedure, this is a veterinary treatment which therefore requires a prescription from the responsible veterinarian)

- keeps the **medicine register up to date** (name of the person responsible for the procedure in which the medicine is used, name of the medicine, batch number, identification of the animals concerned, indication of the treatment)

The 3 main components of the law :

2. The establishments

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• Livret de compétences • Attestations de formation • Diplômes	• Comptes-rendus des réunions de la cellule bien-être animal • Registres entrées/sorties des animaux • Registre entrées/sorties du personnel expérimentateur • Registre des médicaments • Registre des conditions d'environnement (température, hygrométrie...) • Contrôles quotidiens des animaux	• Notification d'autorisation de projet • Affichage des points limites
Inspections		
Visite d'inspection par un inspecteur vétérinaire de la DD(SC)PP (Ministère de l'Agriculture) programmée ou inopinée: inspection d'au moins 1/3 des EU par an et inspection annuelle pour les EU hébergeant des primates non-humains et des carnivores domestiques Vérification des documents réglementaires Vérification des conditions d'hébergement et de fonctionnement, les normes de soin, formation du personnel (réglementaire, continue, livret de compétence) et autorisation des projets		

Animals

obligation to source supplies from **approved breeders and suppliers**

special procedures for primates raised in captivity

stray animals and wild animals: **prohibited** , except by special permission

additional regulations for wildlife (captive or otherwise), for endangered wildlife, non-human primates, great apes

Accommodation and care

- Arrival and departure of animals → entry/exit register

Entry date, origin, sex, age, identification, procedure or project number, exit date, cause of death, permanent corrections to be kept for 5 years

- Animal identification: individual identification is mandatory for primates, dogs and cats, and livestock.

The methods of identification are either mandated or not by regulations.

They must meet the requirements of animal welfare and the requirements of refinement.

- Monitoring of animals:

when kept alive: veterinary supervision

at the end of the process, prioritize adoption: placement programs

euthanasia in cases of prolonged, irreversible, or lasting suffering → definition of the endpoints

(endpoints must be displayed for each project in the experimental room where the project takes place)

- Daily monitoring is MANDATORY (including weekends!) and monitoring of accommodation conditions → registers
- Cages or enclosures of regulated size, clean and suitable bedding, equipment cleaned and disinfected regularly, sufficient food and water of good quality

Animal facility and equipment

(definition in the decree of February 1, ²⁰¹³)

includes: housing facilities, rooms for specific procedures, surgical and post-operative care units, quarantine facilities, laundry (clean circuit and dirty circuit)

They must:

be adapted to the species (soil, cage size, enrichment, outdoor enclosures...)

be made of resistant, washable, disinfectable materials, suitable for preventing injuries

allow the separation of incompatible species

The animal facility must be **approved** by prefectural decree, which is valid for 6 years and renewable upon written request

CERFA form → visit → approval

The 3 main components of the law :

3. The projects

Directive européenne 2010/63/UE		
Décret 2013-118 et 5 arrêtés de publication		
<u>Personnel compétent</u> • Formation spécifique selon fonction exercée (formation approuvée par la CNEA et agréée par le Ministère de l'Agriculture) : concepteur (formation initiale Bac+5 exigée), applicateur, soigneur • Formation continue : 3 jours tous les 6 ans (formations validées par le responsable du suivi des compétences de l'établissement et contrôlées lors des inspections de la DD(CS)PP) • Formations spécifiques : chirurgie ou autres formations approuvées par la CNEA et agréées par le Ministère de l'Agriculture	<u>Etablissement</u> (fournisseur : EF ou utilisateur : EU) Agrément pour 6 ans délivré arrêté préfectoral après visite des locaux par la DD(SC)PP relevant du Ministère de l'Agriculture (via un dossier CERFA) • Vétérinaire responsable • Cellule bien-être animal • Responsable des médicaments • Responsable du suivi des compétences du personnel • Normes d'hébergement	<u>Projet</u> (comprenant des procédures au-delà du seuil de douleur engendrée par l'introduction d'une aiguille effectuée conformément aux bonnes pratiques vétérinaires) • Autorisé pour un maximum de 5 ans par le Ministère de l'Enseignement Supérieur, de la Recherche et de l'Innovation après avis du comité d'éthique dont relève l'établissement (demande d'autorisation via la plateforme APAFIS) • Appréciation rétrospective pour les procédures sévères et procédures sur primates non-humains
• Livret de compétences • Attestations de formation • Diplômes	• Comptes-rendus des réunions de la cellule bien-être animal • Registres entrées/sorties des animaux • Registre entrées/sorties du personnel expérimentateur • Registre des médicaments • Registre des conditions d'environnement (température, hygrométrie...) • Contrôles quotidiens des animaux	• Notification d'autorisation de projet • Affichage des points limites
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Experimental procedures and projects: do not confuse

Project = a work program with a defined scientific objective, which uses one or more animal models, and involves one or more experimental procedures

Procedure (experimental) = any **invasive or non-invasive use** of an animal for experimental or other scientific purposes, or for educational purposes, or any intervention intended or likely to result in the birth or hatching of an animal and the preservation of a line of genetically modified animals

Two project categories : **with** authorization or **without** authorization

Definition : any practice likely to cause greater or equal pain, suffering, or anguish to those caused by the introduction of a needle performed according to good veterinary practice is considered as **a project requiring authorization**.

- Below the threshold: no project authorization

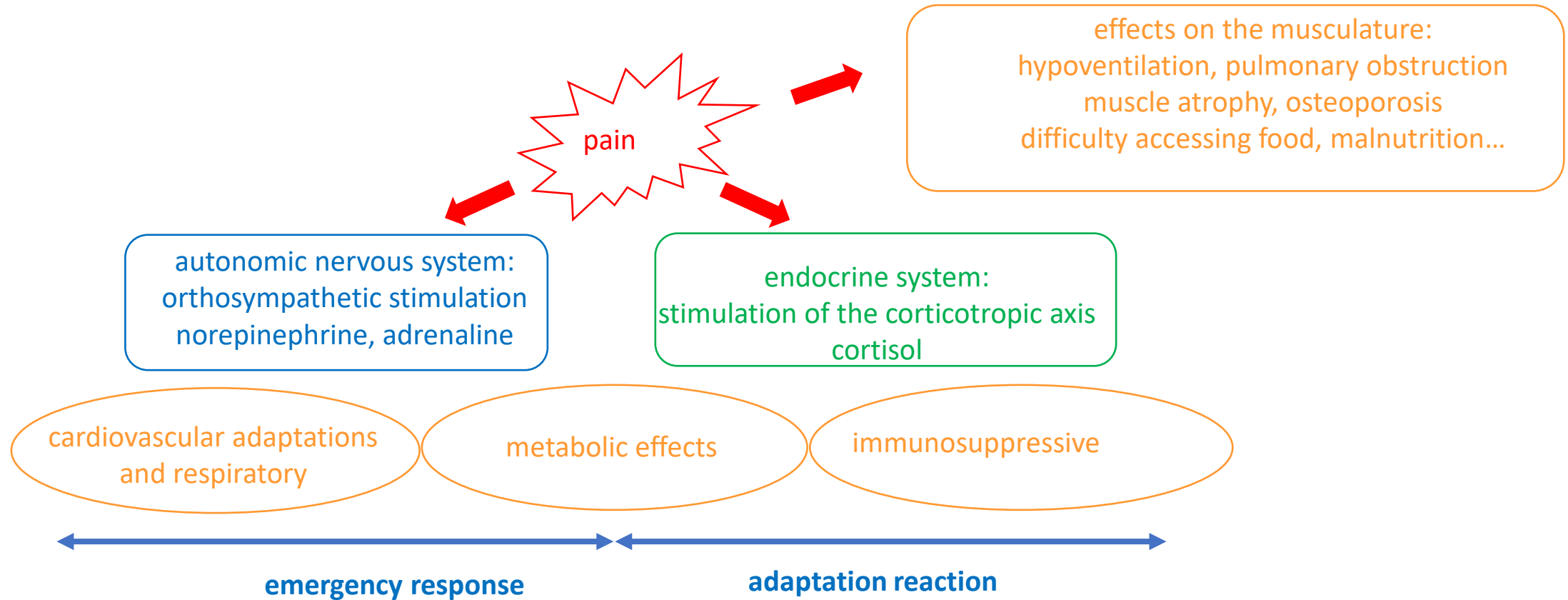
However, a register of projects must be presented during the DDPP visit.

- Killing by one of the authorized methods **is not** a procedure.

**Projects can only be implemented when they are AUTHORIZED
by the Ministry of Higher Education and Research**

For ethical experimentation: pain assessment, establishment of endpoints, and determining the severity of procedures

Pain has physiological and behavioral consequences:



Reminder: Ethical principles of experimentation

*The implementation of proper experimentation must **preserve the animal's welfare** : it is necessary to **recognize the signs of pain** in the species concerned and to **define pain thresholds** to take care of it and, if necessary, beyond which the pain is considered to be no longer bearable.*

This means: evaluating the procedure (intensity, duration, sources) in order to propose management measures

- The concept of **constraint** : a factor that impairs quality of life
→ need to define **indicators of well-being**
- absence of hunger and thirst, physical comfort, good health, possibility of expressing normal patterns of behavior, absence of fear, distress, injury and pain.

+ What are the sources of distress and/or pain?

transportation

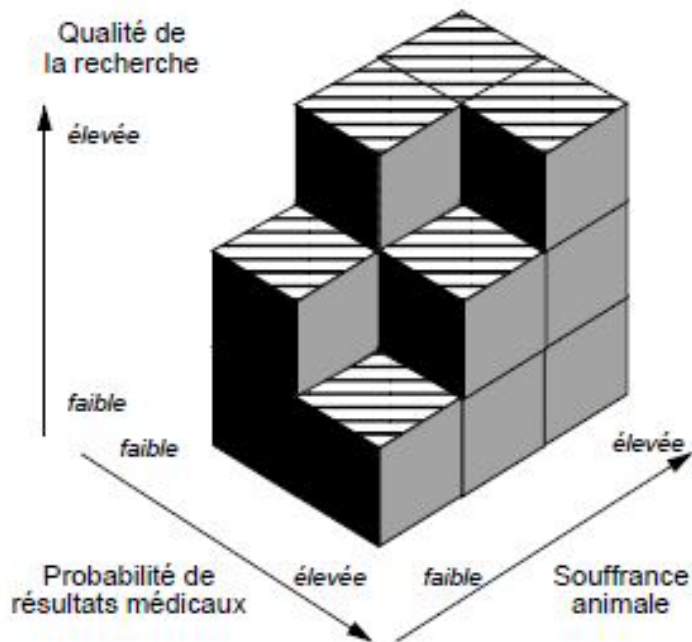
environmental parameters (noise, light, humidity...)

micro-environment (change frequency, re-allotment, etc.)

experimental procedures

Two concepts that must be balanced : achieving the scientific objective / ending suffering

Figure 1. *Appréciation tridimensionnelle de la recevabilité d'une expérimentation (d'après Bateson 1986). Une proposition de recherche qui tombe dans les parties matérialisées du cube doit être rejetée.*



(Veissier, 1999)

Guiding principle :

In experiments conducted on animals, any actual or potential suffering, distress, or discomfort should be minimized or alleviated **by choosing the earliest possible endpoint** compatible with the scientific research objectives.

Two important points to note:

- The handling, restraint, and procedures commonly used can be a source of stress: animals must be accustomed to them before any experimentation (handling, habituation, etc.).
- Stimuli that were not painful in the naive animal may become painful during the experimental procedure

Step 1: Recognizing pain and distress in animals

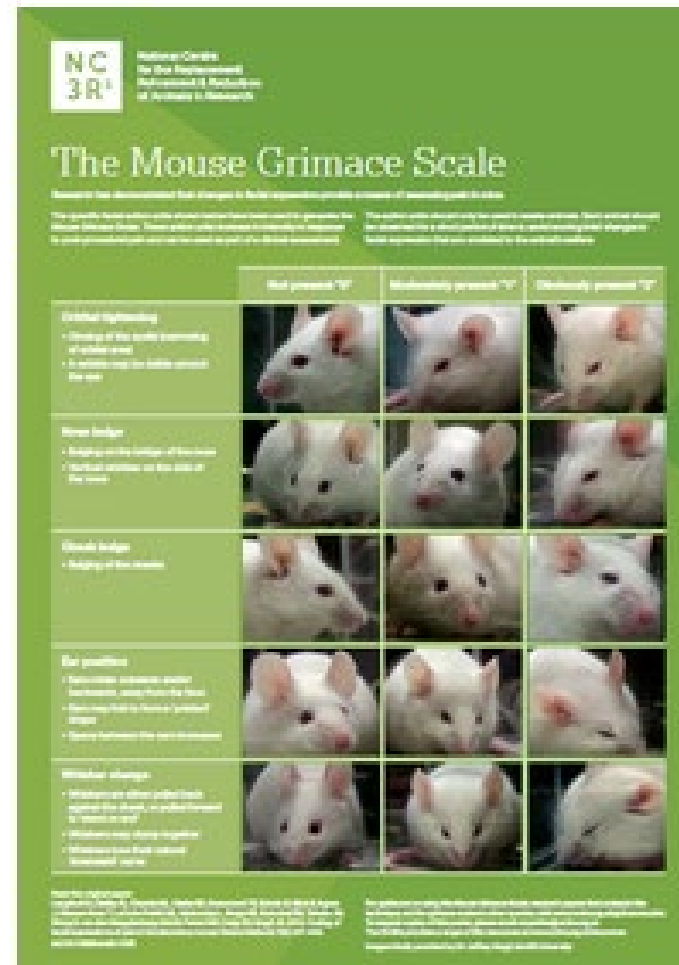
Pain leads to various **behavioral changes** in the animal: learning to avoid repetition of the painful stimulus, protective reflexes (withdrawal), adoption of an antalgic posture to minimize the intensity of the stimulus, aggression (to neutralize the source of the stimulus)

→ need to know the normal behaviors of animals

→ understand the situational, behavioral and physiological indicators of stress, pain and distress.

Main signs of pain in mice

Eyes	Eyelids wide open, half-closed or completely closed,
Breathing	High respiratory rate
Exterior appearance	Ruffled coat, weight loss, dehydration ,
Defecation/urination	Constant stress can decrease or increase defecation or urination
Behavior	Isolation, indifference , aggressive , tendency to
Reflexes	Miosis, mydriasis, muscle fibrillation , absence of reaction
Abnormal activity	Amplified defensive behavior, without reaction to stimuli
Position	Leaks light ,
Locomotion	Cautious
Vibrisses	Increased mobility through stress



Step 2: Define boundary points

Defining a endpoint means **determining how to minimize or reduce** the animal's pain or distress, or **the time** at which it is stopped:

- stopping the process that causes suffering.
- administration of a treatment to relieve pain: analgesia, treatment, modification of the environment.
- euthanasia

For each procedure, it is necessary to :

determine **early** endpoints compatible with the protocol's objective

have a **quantitative assessment** of the animals

determine the **type of intervention** (withdrawal, administration of therapeutic substances, euthanasia)

identify the **actors** (who observes, who reports, who decides)

determine the **frequency** and method of monitoring (in writing in the protocol)

record the observations in writing

record the decisions in writing

Reminder : from a regulatory point of view:

supervision is required **, including weekends!**

The endpoints are based on **parametric signs** (measurements) and on **qualitative signs** (appearance):

- physical appearance
- changes in behavior
- weight fluctuations, body condition
- clinical signs (heart rate, respiratory rate)
- response to stimuli
- tumor size...

The **ideal endpoint** is:

- objective
- predictive varying proportionally to suffering
- quickly measurable and easy to track
- reproducible, reliable and robust

It must be quantified : size, score, hormone levels (catechumen, corticosterone, etc.)...

A **pilot study** can help to define them more precisely

Example of a grid of score clinic – Morton & Griffith 1985

PARAMÈTRE	N° ANIMAL :	NOTE	DATE/ HEURE	DATE/ HEURE
APPARENCE	Normale	0		
	Manque de toilettage	1		
	Pelage terne, écoulements nasaux et lacrymaux	2		
	Piloérection, dos arrondi	3		
ABREUVEMENT ET ALIMENTATION	Normal	0		
	Incertain : perte de poids < 5%	1		
	Diminution : perte de poids 0-15%	2		
	Aucune prise d'aliment ou d'eau	3		
EXAMEN CLINIQUE	T, FC et FR normales	0		
	Faibles modifications	1		
	T+/- 1°C, FC et FR +30%	2		
	T+/-2°C, FC et FR +50% ou très faible	3		
COMPOTEMENT NATUREL	Normal	0		
	Légères modifications	1		
	Moins mobile et attentif, isolé	2		
	Vocalisations spontanées, automutilation, animal agité, ou immobile.	3		
COMPOTEMENT PROVOQUÉ	Normal	0		
	Dépression mineure ou réponse exagérée	1		
	Changements modérés par rapport à ce qui est attendu	2		
	Réaction violente ou très faible et pré-comateuse	3		
AJUSTEMENT DU SCORE	Si un score 3 est noté plus d'une fois, ajouter un point pour chaque 3	2-5		
	Total	0-20		

- 0-4 Normal
- 5-9 Observer attentivement, envisager l'utilisation d'antalgiques
- 10-14 Souffrance, prévoir un mode de soulagement et observer régulièrement. Demander un deuxième avis. Considérer l'euthanasie.
- 15-20 Souffrance sévère. Le protocole expérimental ne doit-il pas être modifié ?

Step 3: Determine the severity of a procedure

The question of pain management and the definition of the limits of each procedure leads to questions about the **degree of severity of the procedure** implemented.

This analysis decides whether or not a **project authorization request is required**.

Regulations:

Annex VI of the decree of 1 February 2013

relating to the ethical assessment and authorization of projects involving the use of animals in experimental procedures

4 levels of severity for the procedures:

Without awakening	procedures experimental done under general anesthesia , and the animal does not regain consciousness at the end of the procedure Attention : it is different to procedures of terminal sampling
Light	pain, suffering or anguish mild and short-lived no significant impact on well-being or general condition
Moderate	pain, suffering or anguish moderate and short-lived Or pain, suffering or anguish light and long-lived moderate impact moderate on the well - being or the general state of the animal
Severe	pain, suffering or anguish intense pain suffering or anguish moderate and long-ivd severe impact moderate on the well - being or the general state of the animal

Level of pain or anguish

Authorization required

Severe

Moderate

Light

Classes of
gravity
procedures

Without wakefulness

Project not requiring authorization

Superior limit = pain, suffering or anguish
intense, likely to be prolonged and impossible to relieve

= > Rural Code, article R .214 - 1 08

*"An experimental procedure is not implemented if it involves intense pain,
suffering, or distress that is likely to be prolonged without the possibility of relief."*



Refinement

Threshold :
introduction
of a needle

Step 4: Implement a refinement strategy

Adapting technical practises



Adapting accommodation



Implementing pain management

2 Description des paliers de douleur prévisibles

Palier 0	Palier 1	Palier 2	Palier 3
ABSENCE	LÉGÈRE	MODÉRÉE	SÉVÈRE
Exemples : biopsie cutanée, prise de sang rétroorbitaire, chirurgie cutanée peu délabrante (voie d'abord pour cathétérisme), injection de produits faiblement irritants	Exemples : laparotomie exploratrice, délabrement tissulaire modéré, injection de produits fortement irritants	Exemples : chirurgie entraînant des troubles persistants, délabrements tissulaires importants	

3 Schéma thérapeutique

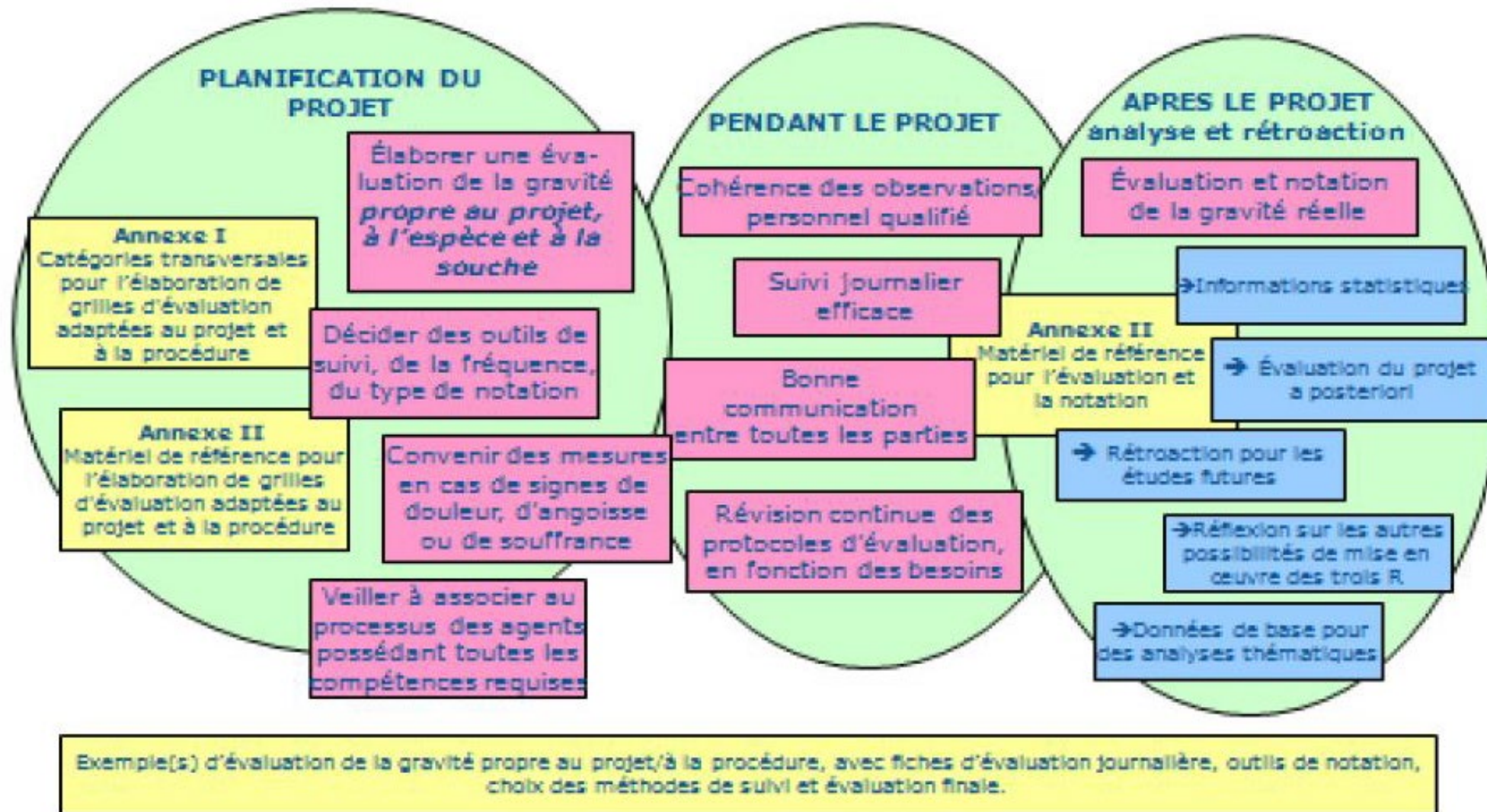
Le choix des méthodes d'analgésie et des molécules est vaste. Il doit être adaptée à chaque cas et à chaque espèce. L'analgésie sera si possible préventive sinon curative et **régulièrement ajustée** en fonction des observations renouvelées sur l'animal.

Palier 0	Palier 1	Palier 2	Palier 3
	AINS (ou autre analgésique faible) ou Morphinique dose faible	Morphinique dose faible à moyenne +/- AINS +/- Anesthésique local	Morphinique dose moyenne à forte +/- AINS +/- Anesthésique local
Exemples d'AINS : kétoprofène, méloxicam, carprofène, phénylbutazone, acide acétyl salicylique, et agents analgésiques faibles : paracétamol (contre-indiqué chez le chat)	Exemples de morphiniques : morphine, buprénorphine, fentanyl, mépéridine	Exemples d'anesthésiques locaux : lidocaïne, bupivacaine	

Remerciements :
Comité d'Éthique, Service d'Anesthésie-Analgésie-Réanimation et Bureau de la Communication de l'École Nationale Vétérinaire de Lyon
Diffusion avec l'aide du CNRS



Severity assessment: a continuous process, for an ethical, scientifically valid and usable experimentation



At the end of the experimental project, what happens to the animals?

Contrairement à ce que l'on pourrait croire, les animaux utilisés à des fins scientifiques **ne sont pas toujours mis à mort** en fin d'étude.

Un animal utilisé à des fins scientifiques a trois devenir possibles :



MISE À MORT



RÉUTILISATION
dans une autre
étude
expérimentale



PLACEMENT

Le devenir de l'animal est décidé en fonction :

- de son **ÉTAT DE SANTÉ**,
- du risque pour la **SANTÉ PUBLIQUE**, la **SANTÉ ANIMALE** ou **L'ENVIRONNEMENT**,
- des différents **CRITÈRES ÉTHIQUES**,
- des **CAPACITÉS** et **SOLUTIONS DE REMPLACEMENT**.

Le placement des animaux utilisés à des fins scientifiques en foyer est permis par la réglementation. Ce principe est parfois considéré comme un **4^{ème} R** de la règle des 3R.

REMPACEMENT

L'utilisation d'animaux doit être remplacée par d'autres méthodes lorsqu'elles existent.

1

RÉDUCTION

Le nombre d'animaux utilisés pour chaque projet doit être réduit au maximum.

2

RAFFINEMENT

Les pratiques mises en place doivent permettre de réduire le plus possible toute douleur, angoisse ou dommages durables que pourraient ressentir les animaux.

3

REPLACE- MENT / RETRAITE

4?

Afin que le placement se passe dans de bonnes conditions, la préparation de l'animal est une étape très importante. Certains laboratoires sont **autonomes** dans ce processus de placement, tandis que d'autres peuvent faire appel à des **associations** (par exemple, le GRAAL).

Placement programs (Article R214-112, amended by Decree 2017-1246 of August 7, 2017 - art. 12)

The retirement of laboratory animals should be considered:

- when their euthanasia is not required for scientific, regulatory or reasons related to the preservation of their general condition and well-being;
- when it is not useful or necessary to keep them in the establishment for scientific or regulatory reasons;
- when there is no intention to reuse them.

needs veterinarian's approval
and animal adaptation to its new environment

Euthanasia/killing

When :

- tissues are needed for analysis, for *in vitro methods*
- surplus of animals (livestock)
- end-of-studies sacrifice
- anticipated sacrifice for humanitarian reasons (endpoints)
- maintenance of health status

Regulations : "*Animals must be euthanasied using a humane method.*"

The authorized methods are specified in Annex IV of the decree of February 1, 2013.

What is a method of human euthanasia?

*An acceptable method must induce :
a loss of consciousness followed by cardiac or respiratory arrest
and a definitive and irreversible loss of brain function
without suffering and distress.*

From the animal's point of view:

painless
as little invasive as possible
rapid loss of consciousness
minimal stress and anxiety

From the experimenter's point of view:

safety, aesthetics, ease of implementation
repetitiveness
absence of interference with the experiment

In practical terms, this means :

- done in a dedicated area separated from housing rooms
- done quickly, without causing emotional shock, with minimal stress for the animal and not dangerous for the experimenter
- carried out by trained personnel → verification through tutoring and recording in the skills booklet
- not causing experimental bias

3. Tableau



Remarques concernant les animaux/méthodes cryptographiques	Poissons	Amphibiens	Reptiles	Oiseaux	Rongeurs	Lapins	Chiens, chats, furets et renards	Grands mammifères	Primates non humains
Surdose d'anesthésique	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)
Tige perforante			(2)						
Dioxyde de carbone					(3)				
Dislocation cervicale				(4)	(5)	(6)			
Commotion/ Percussion de la boîte crânienne				(7)	(8)	(9)	(10)		
Décapitation				(11)	(12)				
Étourdissement électrique	(13)	(13)		(13)		(13)	(13)	(13)	
Gaz inertes (Ar, N ₂)								(14)	
Abattage par balle au moyen de fusils, d'armes à feu et de munitions appropriées			(15)				(16)	(15)	

Si sédation préalable

Augmentation progressive des doses / Pas NN, Pas foetus

< 1Kg ; si >150g sédation préalable

< 1Kg rongeurs < 5 Kg lapins

A n'utiliser qu'en cas impossibilité d'utiliser d'autres méthodes

(1) JO L 303 du 18.11.2009, p. 1.

Following the euthanasia protocol:

Death must be certain, verified, or confirmed (required in the project authorization!):

Confirmation of permanent circulatory arrest :

cessation of heartbeat, respiratory movements, or absence of reflexes

onset of rigor mortis

brain destruction

neck dislocation

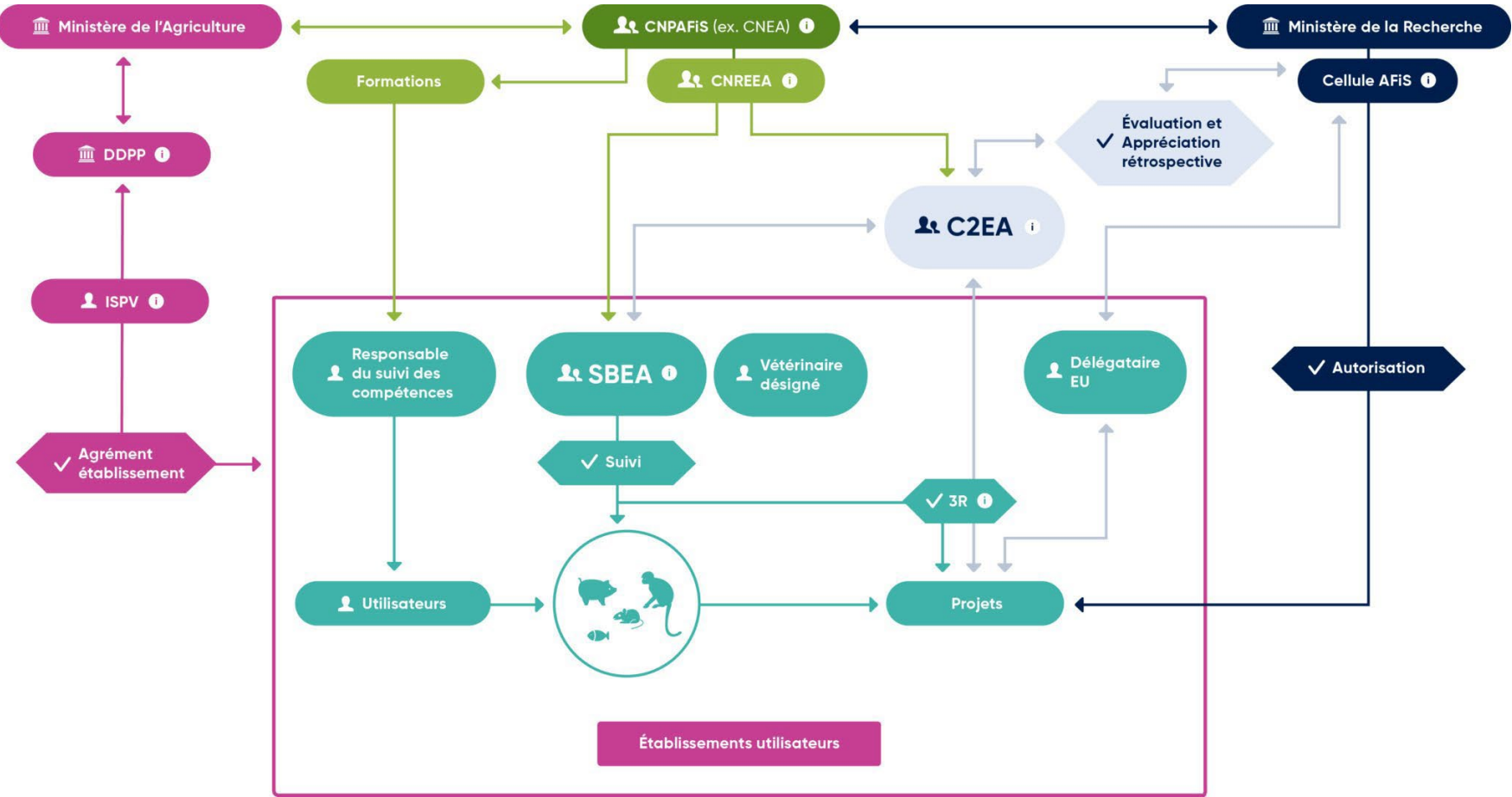
exsanguination

Handling of carcasses

- storage at -20°C in suitable containers, in a closed room, then disposal as infectious medical waste, with prior decontamination if there is a risk of infection.
- if the carcass weighs more than 40 kg: specialized handling by a rendering company
- **traceability** : complete the animal entry/exit register + incineration register (CERFA form)

French institutional framework for ethical oversight and protection of animals used for scientific purposes

<https://www.sbea-c2ea.fr/lanimal-dans-la-recherche/>





Verification that the law is being respected: inspections

- visit for approval or inspection
- by appointment or unscheduled: generally 1/3 of users every year (PNH : every year)
- elements of the inspection

documentation: procedures implemented, project authorizations

staff training

BEA structure: composition, operation, meeting minutes , follow-up on decisions

internal procedures: staffing, animal safety, animal arrival, waste management...

entry/exit registers, medication register

physical: animal housing, condition, behavior, compliance with standards identification

animal care, daily monitoring, enrichment

➤ The inspection results in an inspection report.

➤ For approval: prefectural decree, valid for 6 years

defines the authorized species, the facilities and the responsible staff

Any change must be notified to the prefect (DDPP) by the person in charge of the establishment (species, housing, managers)



**Direction
départementale
de la protection
des populations**



Any visit may be followed by sanctions:

for the establishment

for **individuals** (up to €150,000 in fines and even imprisonment)



CONTRAVENTION DE CLASSE 3

AMENDE MINORÉE

45€

Délai de paiement : 15 jours
(30 jours avec une téléprocédure)

AMENDE FORFAITAIRE

68€

AMENDE MAJORÉE

180€

Délai de paiement : après 45 jours
(après 60 jours avec une téléprocédure)

CONTRAVENTION DE CLASSE 4

AMENDE MINORÉE

90€

Délai de paiement : 15 jours
(30 jours avec une téléprocédure)

AMENDE FORFAITAIRE

135€

AMENDE MAJORÉE

375€

Délai de paiement : après 45 jours
(après 60 jours avec une téléprocédure)



Conclusion

The use of animals for scientific purposes remains **essential** to the production of new knowledge. It uses **sentient living beings** : it can only be practiced within the framework of **rigorous ethics**.

The implementation of this **ethical framework** is guaranteed by:
a **regulatory framework** that everyone must respect
the ongoing assessment of the **severity** of experimental procedures.



III. The regulatory framework for clinical research

Why include human beings in research protocols?

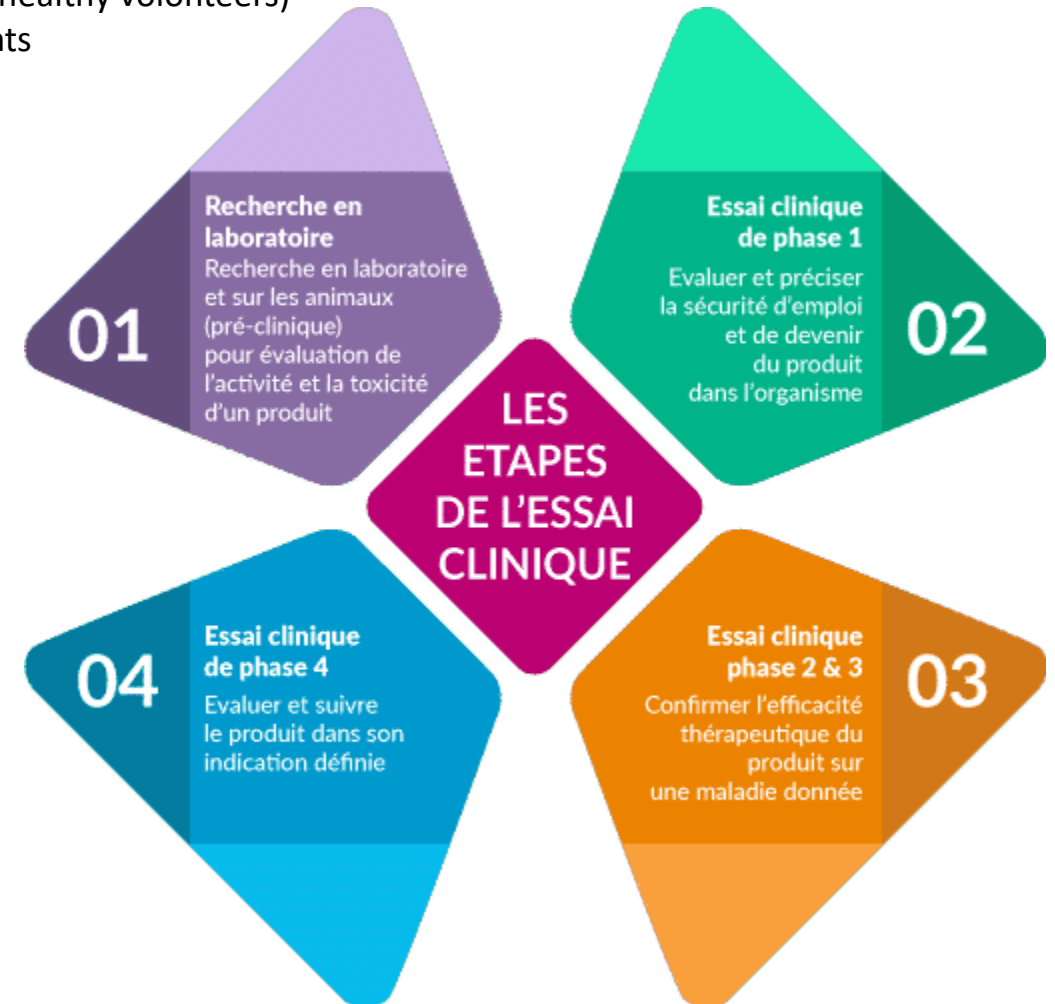
In the context of bringing new medicines, new medical devices, new medical practices, etc....

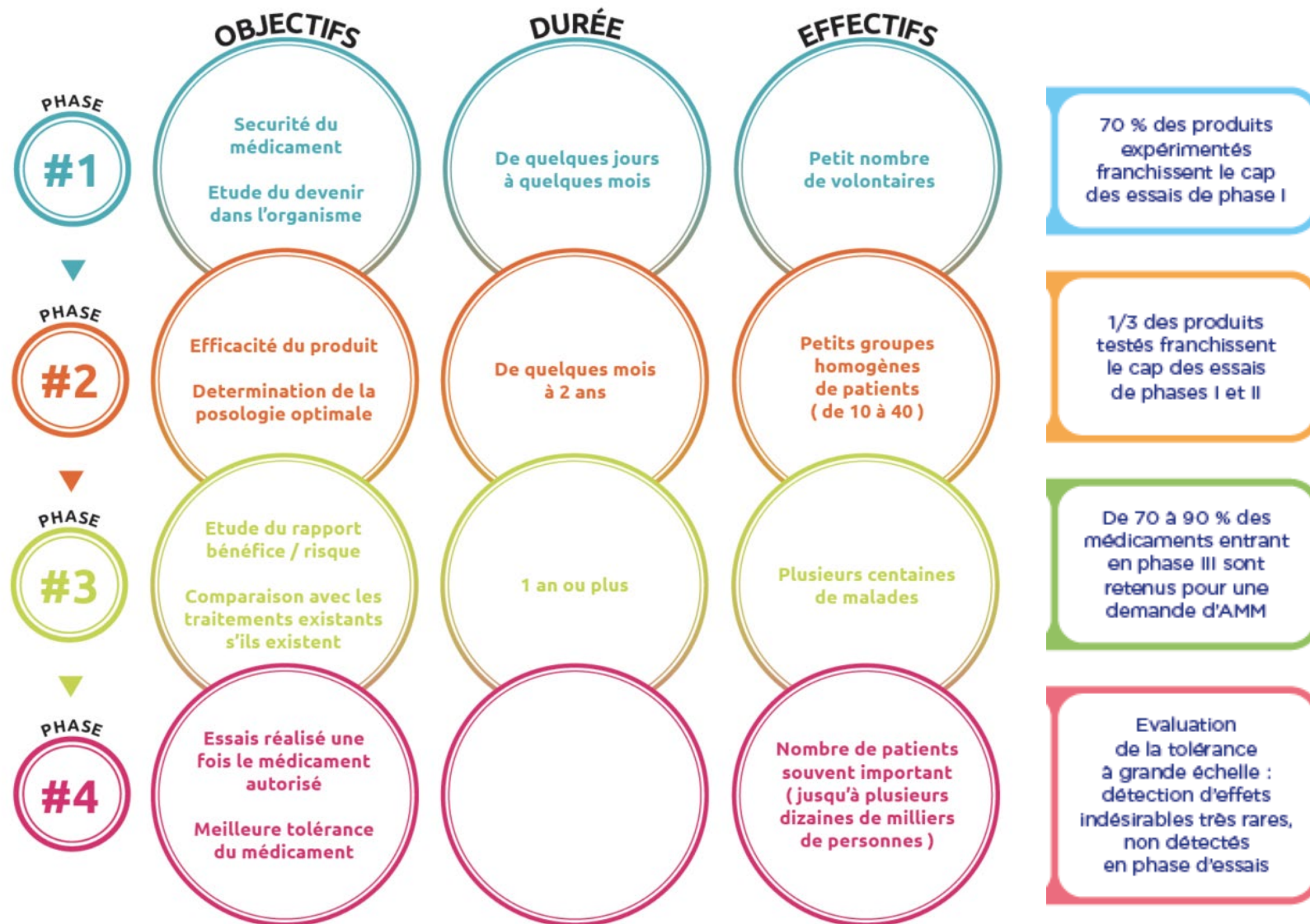
4 successive phases in clinical trials

phase 1 in the laboratory: in silico, in vitro, ex vivo, in vivo (animals and healthy volunteers)

phases 2 and 3 on human subjects: healthy volunteers, volunteer patients

phase 4: post-use monitoring





A brief history of the regulation of research on humans

International Ethical Foundations (1947-1964)

Nuremberg Code (1947) : absolutely essential to receive the informed and voluntary **consent of the subject**

Declaration of Helsinki (1964) principles of test design and scientific validation adopted by the World Medical Association

in Europe and France

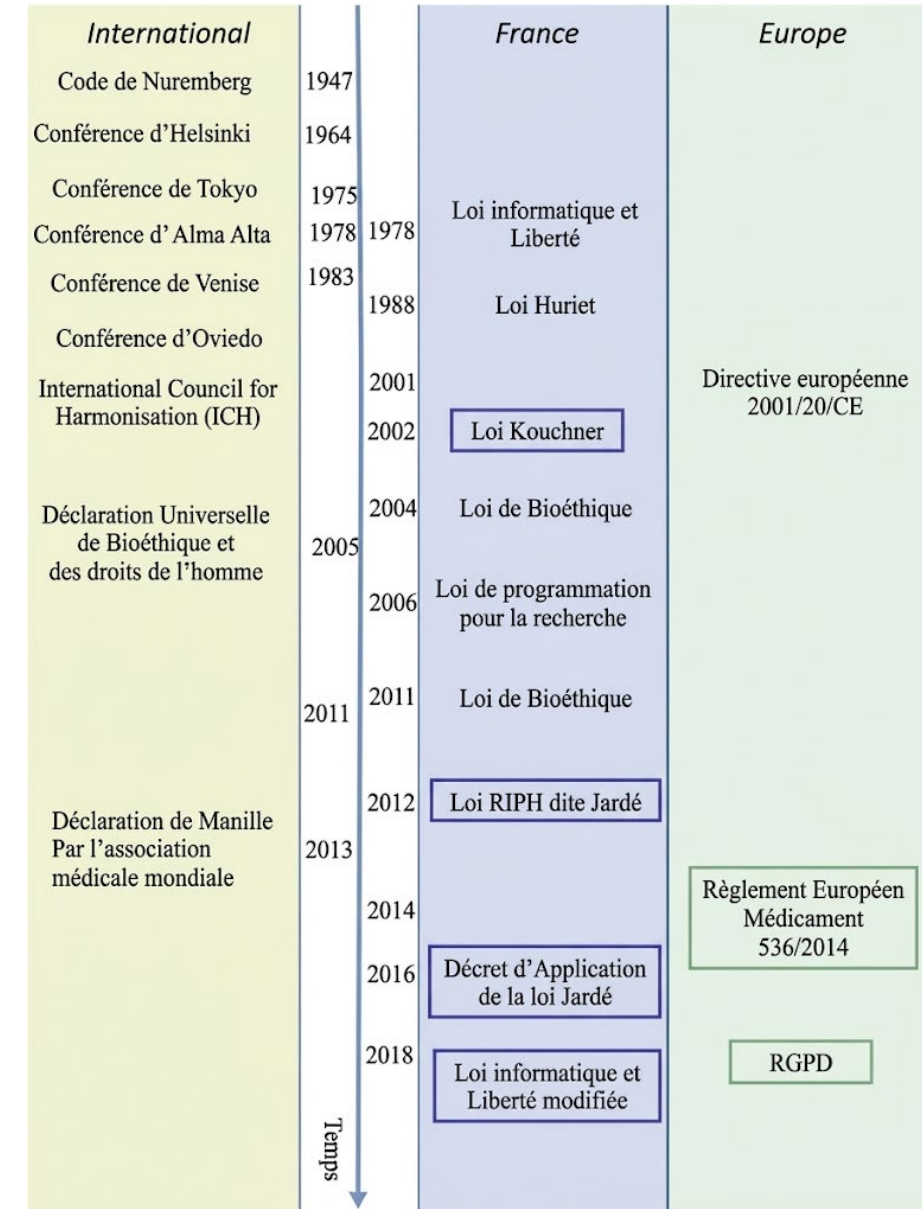
Clinical data for the MA (1975) : Marketing Authorization (MA) dossier for medicinal products requires clinical data based on randomized trials

Sérusclat Law (1988; Law No. 88-1138): relating to the protection of persons participating in biomedical research
legal framework for practices

European Directive (2001) : first European directive relating to clinical trials
transposed to France in 2006

Jardé Law (2012 / implemented in 2016) : redefines research involving human subjects (RIPH)

European Regulation 536/2014 (entered into force in 2022) : imposes a unified methodology for the evaluation and supervision of drug trials in the European Union through the centralized CTIS portal



What does the law currently say?

- 1) Define the search type to comply with applicable rules
 - Implications for the people involved (Jardé law)?
 - Interventional research?
 - Scope of the research?
- 2) In all cases
 - Patient information and consent
 - Justification for compliance with the protection of the data of the persons concerned
- 3) Most often
 - Favorable opinion from an ethics committee
 - Authorization from the competent authorities

Les acteurs de la recherche clinique

Autorités réglementaires

- Agence nationale de sécurité du médicament et des produits de santé (ANSM) ;
- Comité de protection des personnes (CPP) ;
- Commission nationale de l'informatique et des libertés (CNIL).



Participant

Personne volontaire : saine ou malade.



Investigateur et son équipe

- Peut être médecin, pharmacien, kinésithérapeute, etc.
- Il travaille avec une équipe composée notamment d'infirmiers, d'attachés de recherche clinique (ARC), psychologues, etc.

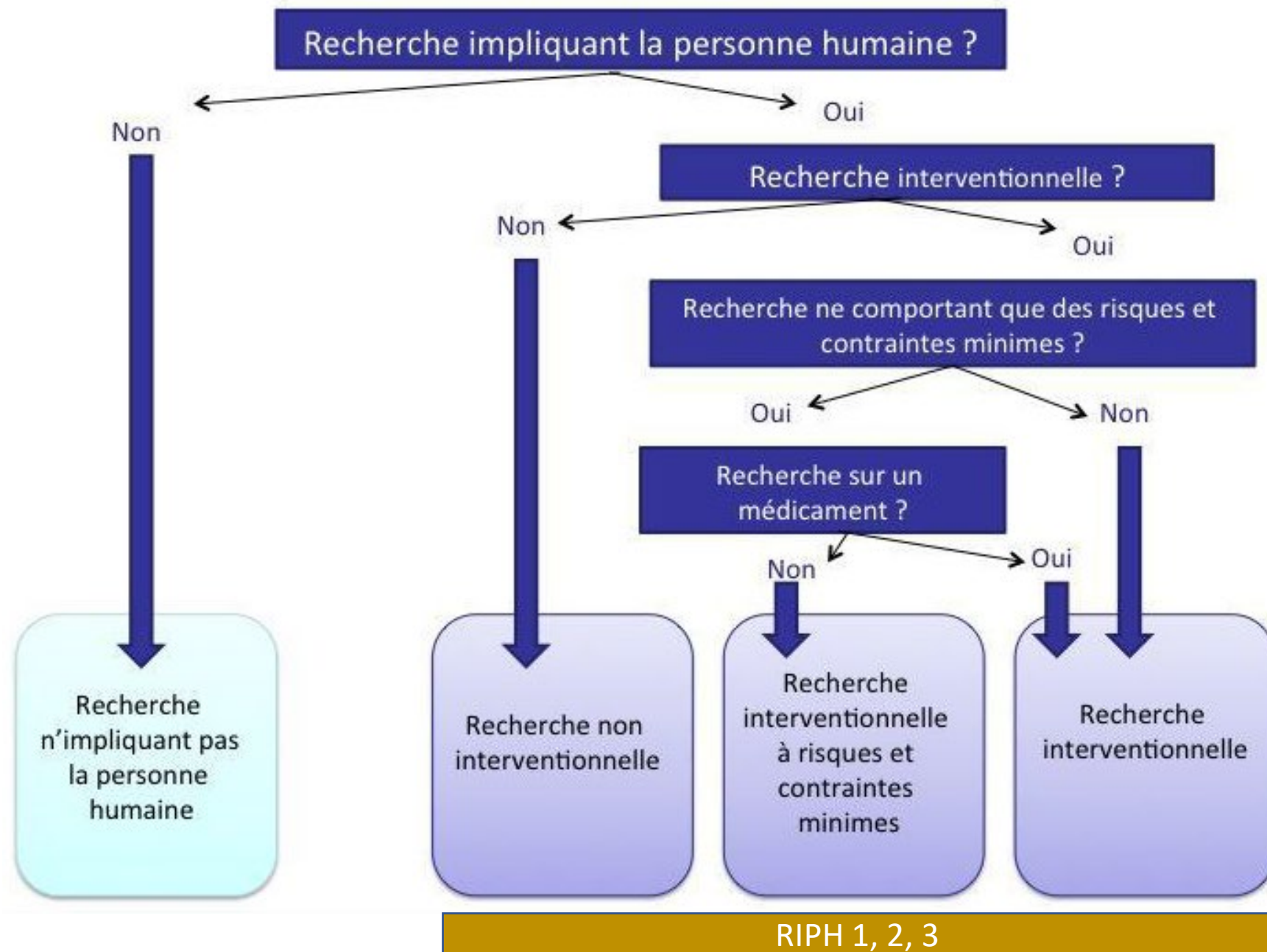


Promoteur

Peut être :

- Personne(s) : médecins, pharmaciens, etc.
- Organisme(s) public(s) : établissement de santé, fondation, etc.
- Organisme(s) privé(s) : laboratoire pharmaceutique, fabricant de dispositif médical, etc.





Research not involving human subjects
(research based on data already collected outside of *RIPH3*)
outside the framework of the Jardé law

patient satisfaction surveys

experiments in the human and social sciences in the field of health

evaluation of the working methods or teaching practices of healthcare professionals

studies on biological data or samples already collected outside of genetic studies

Recherches Impliquant la Personne Humaine (RIPH)

```
graph TD; A[Recherches Impliquant la Personne Humaine (RIPH)] --> B[Recherches Interventionnelles]; A --> C[Recherches non Interventionnelles]; B --> D[RIPH de Catégorie 1]; B --> E[RIPH de Catégorie 2]; C --> F[RIPH de Catégorie 3]; D --> G[Recherches interventionnelles qui comportent une intervention sur la personne non justifiée par sa prise en charge habituelle]; E --> H[Les recherches interventionnelles qui ne comportent que des risques et des contraintes minimales, dont la liste est fixée par arrêté]; F --> I[Les recherches non interventionnelles dans lesquelles tous les actes sont pratiqués et les produits utilisés de manière habituelle, sans procédure supplémentaire ou inhabituelle de diagnostic, de traitement ou de surveillance];
```

Recherches Interventionnelles

RIPH de Catégorie 1

Recherches interventionnelles qui comportent une **intervention sur la personne non justifiée par sa prise en charge habituelle**

RIPH de Catégorie 2

Les recherches interventionnelles qui ne comportent que des **risques et des contraintes minimales**, dont la liste est fixée par **arrêté**

Excluding medications

Recherches non Interventionnelles

RIPH de Catégorie 3

Les recherches non interventionnelles dans lesquelles tous **les actes sont pratiqués et les produits utilisés de manière habituelle**, sans procédure supplémentaire ou inhabituelle de diagnostic, de traitement ou de surveillance

Tableau 1 - Réglementations associées aux 3 catégories de recherches définies par la loi Jardé			
Recherches interventionnelles à risques (art L1121-1-1°)		Recherches interventionnelles à risques minimes (art L1121-1-2°)	Recherches non interventionnelles (art L1121-1-3°)
Recherches sur des médicaments*	Recherches hors médicaments	Recherches hors médicaments définies par l'arrêté du 2 décembre 2016 annexe 1	Recherches observationnelles Produits de santé (médicaments compris) et hors produits de santé
Enregistrement n° eudraCT	Enregistrement n°ID-RCB		
Autorisation ANSM**		Information ANSM	
Avis CPP			
Information Individuelle de la personne			
Consentement écrit	Consentement écrit ou oral		Non-opposition
CNIL : engagement de conformité MR001			CNIL : engagement de conformité MR003
Promoteur			
Assurance			

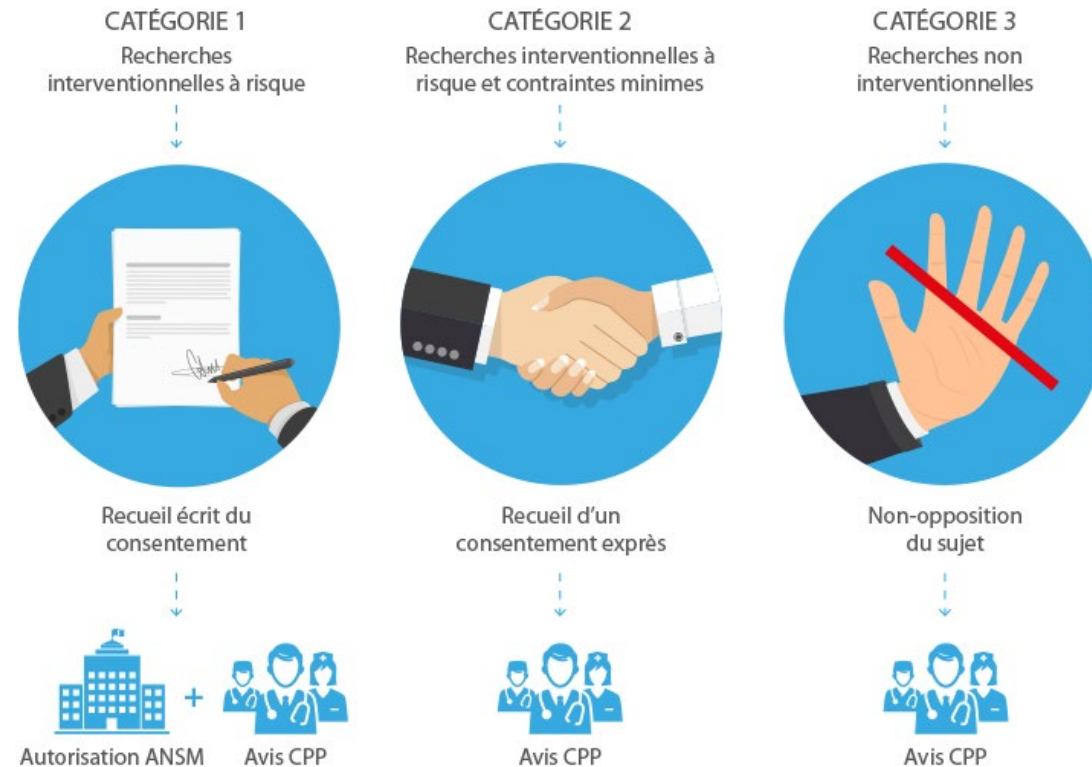
* À partir de 2018, soumis au règlement européen. ** Ou règlement européen pour les médicaments à partir de 2018.

The concept of consent:

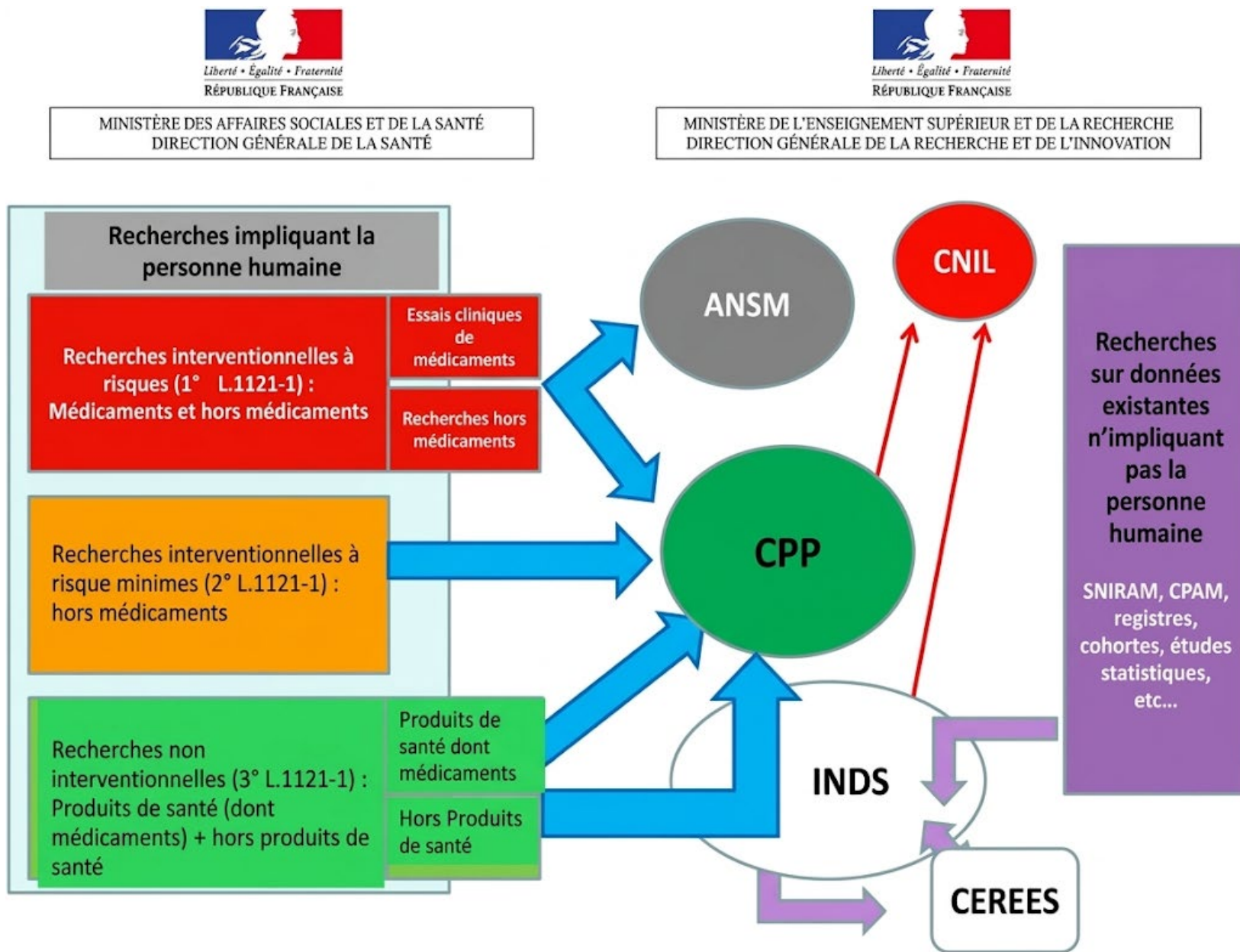
Free, informed and explicit consent given by a person to participate in biomedical research, after having received clear information from the medical team

Key principles:

- free and voluntary: the participant is under no pressure and can refuse or withdraw its consent at any time, without justification or impact on the continuation of their care.
- informed: the patient receives a detailed information sheet describing the objectives, the timeline, the potential benefits, the constraints (appointments, examinations) and the risks or effects
- formalized in writing: signature of a form by the participant (or their legal representative if a minor or protected adult) and the investigating physician
- continuous re-information: if new data emerges during the study, participants must be informed to confirm their willingness to remain in the trial.



The authorities responsible for implementing the law on biomedical research



Ethics committees



Comité consultatif national d'éthique pour les sciences de la vie et de la santé

The Committees for the Protection of Persons (CPP):

required by law for research involving human subjects under the public health code (interventional research)

give a decision-making opinion that is essential for carrying out the tests

are structures approved by the Ministry of Health for a period of six years and endowed with specific territorial competence

they give reasoned opinions before any biomedical research

if the opinion is unfavorable, the search is prohibited.

they guarantee the protection of participants in biomedical research by verifying :

the information provided to patients (prior information, consent, exclusion period, reflection period, etc.)

the relevance of the research, with a satisfactory benefit/risk assessment and an appropriate methodology.



Research Ethics Committees (RECs) / Institutional Review Boards (IRB):

institutional (such as at Inserm or in universities)

evaluate non-interventional research projects (e.g., retrospective studies or studies on health data) which do not strictly fall under CPP but nevertheless require recognized ethical validation for international scientific publication.

Most international journals require an ethical opinion regardless of the type of study → an **IRB provides this ethical opinion for research outside the Jardé law (which does not fall under a CPP).**



Expertise :

- medications : all medications (before and after marketing authorization) and raw materials, including narcotics and psychotropic substances, vaccines, homeopathic products, herbal products and preparations, compounded and hospital preparations.
- biological products: organs, tissues and cells used for therapeutic purposes; cell and gene therapy products; labile blood products; ancillary therapeutic products
- medical devices: therapeutic, diagnostic, in vitro diagnostic, technical platforms, medical software
- other health products, cosmetics and tattoo products

Public health police action :

- scientific and technical evaluation of the quality, efficacy and safety of medications and biological products
- continuous monitoring of adverse effects of health products
- inspection of establishments involved in manufacturing, importing, distributing, pharmacovigilance and conducting clinical trials
- drug and biological product control, product withdrawal/release
- authorizations: marketing authorizations (including withdrawal or suspension), import authorizations, clinical trial authorizations, bans/withdrawals
- information (patients, healthcare professionals, learned societies)

Conclusion

15 grandes dates de la médecine

Le médecin anglais Edward Jenner réussit la première vaccination (contre la variole, une maladie très contagieuse).

1796

1628 Le médecin anglais William Harvey découvre la circulation du sang. Vers 1660, le Français Jean Pecquet découvre la circulation de la lymphe (liquide transparent dans lequel baignent les cellules du corps).

1628



Le dentiste américain William Morton anesthésie l'un de ses patients avec de l'éther (liquide à l'odeur forte), avant une opération.

1846

Le médecin allemand Robert Koch découvre la bactérie responsable de la tuberculose, alors considérée comme la maladie du siècle.

1882

Le médecin français Louis Pasteur invente le vaccin contre la rage (une maladie mortelle transmise par la morsure d'un animal).

1885



Le savant allemand Wilhelm Conrad Röntgen réalise la première radio.

1895

L'Américain Jonas Salk découvre le vaccin contre la poliomyélite, une maladie très grave qui touche surtout les jeunes enfants.

1952

Le médecin sud-africain Christiaan Barnard réalise la première greffe de cœur. Une greffe est une opération consistant à remplacer un organe malade par un organe sain, pris sur un autre être vivant.

1967

L'équipe du professeur français Luc Montagnier identifie le VIH, responsable du sida, une grave maladie (qui se transmet par le sang, lors de rapports sexuels...).

1983

Des médecins français et belges réussissent, à l'hôpital d'Amiens (Somme), la première greffe de visage sur une femme. Elle avait été gravement défigurée après avoir été mordue par un chien.

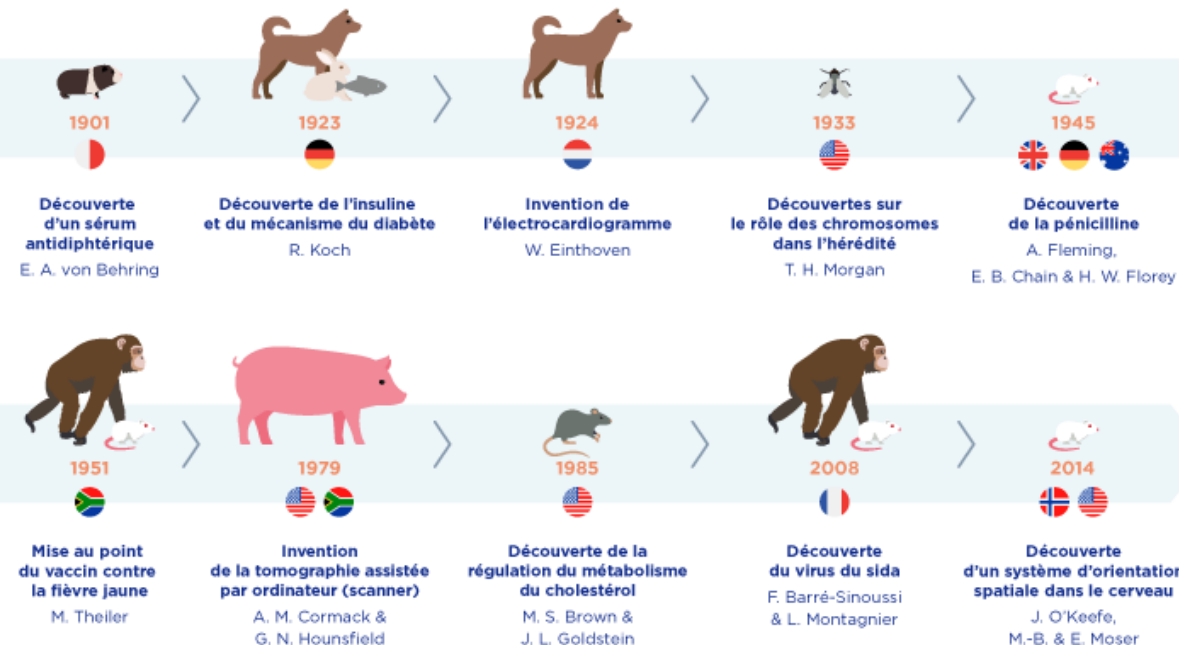
2005

1996 2 scientifiques britanniques réussissent à cloner (faire une copie exacte) une brebis : l'animal est nommé Dolly.

1996



LES GRANDES DÉCOUVERTES LIÉES À LA RECHERCHE ANIMALE



Source : <https://www.recherche-animale.org/1es-priv-nobel>

